Discriminating Because Others Might: Anticipating Third-Party Discrimination

Nicholas C. Owsley*

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Abstract

Research on discrimination typically focuses on decision makers' own unfavorable beliefs or preferences toward marginalized groups. Yet decision makers may also anticipate bias from third parties—clients, coworkers, or shareholders—and discriminate in turn, even when they hold no such bias themselves. In two preregistered online experiments, I show that beliefs about third-party discrimination are a substantial driver of decision makers' own hiring choices. In a natural experiment leveraging the 2024 U.S. presidential election, hiring managers in states where Donald Trump won the most votes hired fewer female workers after the election than before—a change that did not occur for managers in states where Kamala Harris won. This shift was explained by post-election changes in how much discrimination managers expected from *clients* in their states. In a controlled experiment, exogenously reducing managers' expectations about client discrimination increased their hiring of female workers. Across both studies, decision makers systematically overestimated third-party discrimination, amplifying the penalty they imposed on female workers. Together, these results identify beliefs about third parties as an important driver of discrimination, independent of directly held beliefs about stereotyped groups, and show that large-scale social events like elections can shift discrimination by changing these beliefs.

*University of Chicago, Booth School of Business, nowsley@chicagobooth.edu; orcid: 0000-0001-7898-6947

1 Introduction

Discrimination based on gender, race or other immutable characteristics occurs across a range of organizational contexts and contributes to persistent gaps in hiring, promotion, and wages for women and racial minorities (Bertrand & Duflo, 2017; Gaddis, Larsen, Crabtree, & Holbein, 2021; Hangartner, Kopp, & Siegenthaler, 2021; Kline, Rose, & Walters, 2022). Existing research has primarily focused on two channels: discrimination driven by preferences about specific groups (i.e., taste-based discrimination) and discrimination driven by beliefs about group members' attributes, such as their ability (i.e., statistical discrimination).[1] This paper considers the possibility that people discriminate not because of their own preferences or beliefs about the characteristics of some group, but based on beliefs about discrimination by *others*. For example, a business owner might not hire Black workers if he thinks customers won't frequent his business when they see a Black employee behind the counter instead of a White one. An executive recruitment firm might not propose a woman candidate for CEO if they believe shareholders are prejudiced against female executives. A hiring manager may not hire women because he believes their supervisor will invest less in their development than similar male hires. While previous work suggests such "third-party" discrimination may play an important role in labor markets (Abraham, 2020; Becker, 1993; Fernandez-Mateo & Fernandez, 2016; Holzer & Ihlanfeldt, 1998; Hurst, Rubinstein, & Shimizu, 2024; Leonard, Levine, & Giuliano, 2010; Neumark, Bank, & Van Nort, 1996), there is little direct causal evidence of its existence or significance.

Beliefs about third-party actions have been shown to affect a range of important behaviors, such as financial investing, energy consumption, alcohol consumption, and domestic violence (Jachimowicz, Hauser, O'Brien, Sherman, & Galinsky, 2018; Tankard & Paluck, 2016). This occurs both because beliefs that third parties engage in a behavior might make that behavior "normative," encouraging norm-consistent behavior, and because third-party behaviors can directly affect decision makers' payoffs, incentivizing them to anticipate and account for those behaviors (Chang, Milkman, Chugh, & Akinola, 2019; Cialdini & Goldstein, 2004; Tankard & Paluck, 2016). In organizations, for

[1] See Guryan and Charles (2013), and see Bohren, Haggag, Imas, and Pope (2023) for a recent review.

example, if clients, shareholders, or employees discriminate against members of a certain group (e.g., women), hiring or promoting members of that group could be costly — through reduced sales, profits, share price, or other outcomes (Kelley, Lane, Pecenco, & Rubin, 2024; Leonard et al., 2010). A group member's expected value to the organization thus depends not only on their own attributes, but also on how the decision maker expects others to treat them. Beliefs about third-party discrimination can therefore generate a distinct form of statistical discrimination—one that requires no negative beliefs about the group itself. However, it remains unclear whether organizational decision makers anticipate such third-party discrimination, what factors affect these beliefs, and how these beliefs translate into labor market outcomes for marginalized groups.

The present work investigates these questions by both directly manipulating third-party beliefs in a controlled experiment and by leveraging a natural shock to these beliefs. This paper also focuses on the *accuracy* of these beliefs. It would be one thing if people acted on accurate beliefs about third-party discrimination—preferring candidates from a certain group because it would be costly to do otherwise—but beliefs about third-party discrimination can also be inaccurate. Indeed, research on social norms and pluralistic ignorance suggests that beliefs about others' attitudes, beliefs and behaviors can be systematically wrong (Bursztyn, Cappelen, Tungodden, Voena, & Yanagizawa-Drott, 2023; Prentice & Miller, 1996). People tend to have incomplete information about what others do or think, which is likely to be more pronounced for sensitive topics like discrimination (Tankard & Paluck, 2017). Research on the "bias blind spot" effect also shows that people typically overestimate beliefs and actions considered "biased" (Pronin, Gilovich, & Ross, 2004; Pronin, Lin, & Ross, 2002; Scopelliti et al., 2015). Bursztyn et al. (2023), for example, found that people systematically overestimated opposition to female labor force participation. If decision makers are upwardly biased in their beliefs about third-party discrimination, then acting on these beliefs might exacerbate, rather than simply mirror or reproduce existing discrimination. In fact, if no such discrimination by third parties exists, but decision makers believe it does, this could lead to a special form of "self-fulfilling prophecy," whereby discrimination only occurs because of these incorrect beliefs.

In this paper, I test for "third-party discrimination"—whether decision makers discriminate because of their beliefs regarding how *others* discriminate—and highlight the role of inaccurate beliefs in generating a "self-fulfilling prophecy." I also show that aggregators of social information, e.g., results of a national election, affect these third-party beliefs, and demonstrate their downstream consequences on behavior. I investigate these questions in a stylized online labor market using both a natural experiment that leverages the 2024 U.S. presidential election and a controlled experiment. I first measure and utilize the shift in beliefs about third-party gender discrimination induced by Donald Trump's election victory over Kamala Harris in November 2024. Previous work shows that public signals, like national election results, can affect related third-party beliefs.[2] I propose that learning about a Donald Trump election victory, vis-a-vis a Kamala Harris victory, affected decision makers' beliefs about third-party gender discrimination (I explain the rationale behind this proposition in more detail in Section 2.1.5). This, in turn, could affect their choices to discriminate directly, such as whether to hire a female worker for a client-facing role. To identify the effect of the election on these beliefs and discrimination, I measured beliefs and hiring choices immediately before and after the election with participants from states where the election outcome was ex ante uncertain. I then exploited variation in both which candidate won the election across different states and in the "surprisingness" of the result to individual participants within states, testing how this predicted changes in third-party beliefs and hiring choices over the election period.

I find that the election meaningfully shifted participants' beliefs about gender discrimination by others in their state: in states where Trump won, hiring managers' beliefs about the extent of client discrimination against female workers significantly increased. Hiring managers in these "Trump States" responded by hiring significantly fewer female workers after the election than before, particularly compared to managers in "Harris States." This result was mostly driven by those who did not anticipate the election outcome in their state. Hiring managers in Trump states who expected Harris to win and were thus "surprised" by a Trump victory, shifted their beliefs regarding client discrimination more than those who accurately predicted a Trump victory, and hired fewer women as a result. This pattern was reversed in states where Harris won the highest vote share.

Several analyses support the interpretation that these changes in beliefs about client discrimination

[2]Large-scale social and political events– like Supreme Court rulings, national elections, and large-scale protests– serve as important aggregators of social information, providing decision makers with information about the beliefs, preferences, and behaviors of others in their society. This has been shown to affect beliefs about third-party behavior, and related behavior towards marginalized groups. See, for example: Bursztyn, Egorov, and Fiorin (2020); Grosjean, Masera, and Yousaf (2023); Larreboure and Gonzalez (2021); Müller and Schwarz (2023); Tankard and Paluck (2016, 2017).

caused the changes in hiring, and rule out the most plausible alternative explanations, such as changes in beliefs about male-female productivity or selection effects. However, to provide more support for the proposed mechanism, I also conducted a controlled online experiment aimed at isolating the effect of third-party beliefs directly. Here, I used the same hiring paradigm, but directly manipulated beliefs about client discrimination, holding other factors fixed. The results confirmed those of the natural experiment—hiring managers responded to signals that clients discriminate less against women by hiring more female workers. Taken together, these studies demonstrate the causal effect of beliefs about third-party discrimination on hiring decisions.

However, the critical question remains: are hiring managers' beliefs about client discrimination accurate? In both experiments, I find that hiring managers overestimated clients' discrimination against female workers and consequently hired fewer female workers than they would with accurate beliefs. In fact, clients in both experiments did not discriminate against female workers, while hiring managers believed that they did, and these incorrect beliefs caused them to discriminate. This then produced discrimination in this market as a special form of "self-fulfilling prophecy".

This paper makes several contributions. First, it extends the literature on statistical discrimination (Arrow, 1973; Phelps, 1972), which has traditionally focused on the role of beliefs about group members' own attributes, such as their job-relevant skills (Barron, Ditlmann, Gehrig, & Schweighofer-Kodritsch, 2024; Bohren et al., 2023; Campos-Mercade & Mengel, 2023). The present work broadens the set of beliefs that can generate statistical discrimination to include beliefs about how *others* discriminate. These beliefs can also affect a group member's expected value to the decision maker, but can arise independently of views of the group itself. It also builds on previous research on customer discrimination (Holzer & Ihlanfeldt, 1998; Hurst et al., 2024; Neumark et al., 1996), which shows that firms tend to favor workers with the same demographic characteristics as their clients. The present work directly manipulates beliefs about clients, rather than inferring them from their demographics, and identifies the effect of these beliefs on hiring. In this way, the present paper connects the literature on customer discrimination with that on statistical discrimination by centering third-party beliefs as a mechanism for customer discrimination.

This work also contributes to the literature on belief biases and discrimination. There has been a surge of recent work showing that motivational, perceptual, and inferential biases—such as the representativeness heuristic (Bordalo, Coffman, Gennaioli, & Shleifer, 2016; Coffman, Exley, & Niederle, 2021; Esponda, Oprea, & Yuksel, 2023), non-Bayesian belief updating (Campos-Mercade & Mengel, 2023; Coffman, Collis, & Kulkarni, 2024), and motivated reasoning (Eyting, 2022; Rackstraw, 2022; Stötzer & Zimmermann, 2024)—can cause belief distortions about members of certain groups and discrimination. This paper proposes and demonstrates that a form of "pluralistic ignorance" (Bursztyn et al., 2023; Prentice & Miller, 1996; Pronin et al., 2002), or systematic misperceptions of others' beliefs and behaviors, can also cause biased beliefs and discrimination.

This paper also builds on recent research on prejudice accommodation (Lide & Berg, 2023; Vial, Brescoll, & Dovidio, 2019; Vial, Dovidio, & Brescoll, 2019). This prior work shows that when decision makers receive information suggesting a manager is prejudiced (e.g., they become aware of a manager's personal beliefs or infer such beliefs from the degree of diversity in his firm) they "accommodate" the manager's prejudice by hiring fewer people from the targeted group. The present paper builds on this work by testing how beliefs about discrimination by "general others" (such as a randomly drawn client) can affect candidate selection, and by showing how naturally occurring public signals (like an election) might affect these beliefs. Moreover, by testing the accuracy of these beliefs compared to ground truth, I also show that these beliefs can be systematically wrong and can exacerbate existing discrimination in settings with real monetary stakes.

Finally, this work adds to a growing literature on the effect of landmark social events—such as national elections, Supreme Court decisions, and large-scale protests—on discrimination and other inter-group attitudes and behaviors (Edwards & Rushin, 2018; Feinberg, Branton, & Martinez-Ebers, 2022; Grosjean et al., 2023; Larreboure & Gonzalez, 2021; Müller & Schwarz, 2023). The present work provides evidence of a potential causal channel for the above effects, namely that these events affected decision makers' third-party beliefs, and consequently their own behavior, and extends the above findings to discrimination in the labor market.

2 A Natural Experiment

2.1 Design

To test for the existence and implications of third-party discrimination, I conducted a preregistered incentivized hiring experiment in a stylized online labor market (AsPredicted #197318). In this study, I used the 2024 U.S. presidential election as a natural source of variation in beliefs about third-party discrimination, and examine how those beliefs impacted gender discrimination in hiring. Participants played one of three roles in this labor market: workers (who completed a set of incentivized tasks), clients (who selected workers based on their expected task performance), and hiring managers (who hired workers they believed clients preferred). Hiring managers were specifically paired with clients from their own state of residence for all choices and belief measures. This setup allowed hiring manager and client decisions to be fully incentivized—about real online workers—and free of deception. For clients and hiring managers, I measured worker selection in a hiring task and a set of beliefs twice, both (i) before the 2024 U.S. presidential election (the "baseline survey"; conducted 1-2 days before the election) and (ii) after the election (the "endline survey", conducted 11-15 days after the election). This allowed me to measure how participants' choices and beliefs changed over the election period. I focus my analysis on changes in hiring managers' choices and beliefs about client discrimination. Below I give more detail on each stage of the experiment.

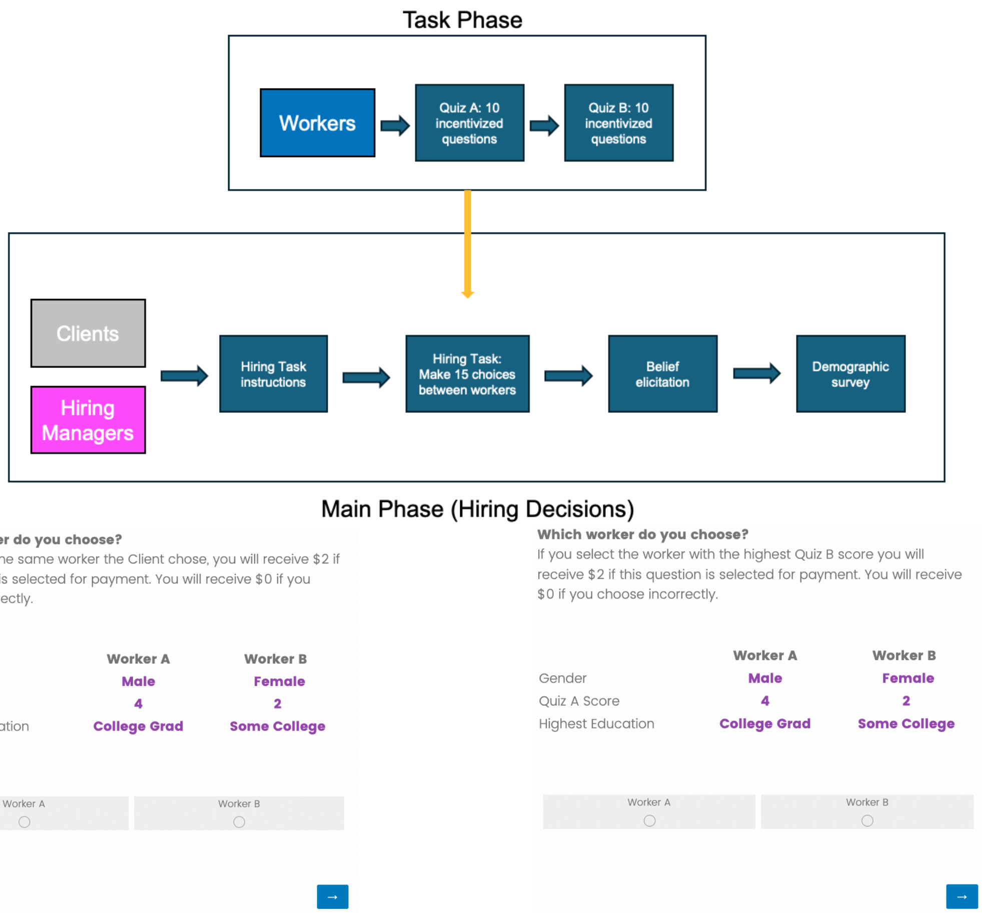
2.1.1 Hiring Task

Figure 1, panel (a) summarizes the survey flow for each participant type. In the first phase of this experiment, the "task phase", 50 male and 50 female online workers completed two incentivized "male-stereotyped" tasks: two quizzes on math, finance and history (called "Quiz A" and "Quiz B"). People typically expect female workers to perform worse than male workers on these and similar tasks, and also expect that others believe this (Bertrand & Duflo, 2017; Bohren, Hull, & Imas, 2022; Coffman, 2014; Dustan, Koutout, & Leo, 2022; Feld, Ip, Leibbrandt, & Vecci, 2022; Guiso, Monte, Sapienza, & Zingales, 2008).[3] In these data, however, male and female workers performed approximately equally well on both tasks (see Figure 6 for the distribution of scores in

[3] In a previous study using this identical task, Prolific participants believed male workers outperformed female workers (Bohren et al., 2022).

Figure 1: Survey Flow and Hiring Task Decision Screens

(a) Survey Flow for Each Phase



(b) Hiring Manager Decision Screen

(c) Client Decision Screen

Notes: Panel (a) shows the experimental flow for the task phase and the main phase, which was the same for both the controlled and natural experiment. Note, that clients and hiring managers completed the same flow for the main phase twice in the natural experiment: both before and after the 2024 U.S. presidential election. Panels (b) and (c) give examples of the decision screens that hiring managers and clients saw during the hiring task, respectively. They each saw 15 decision screens like this, with workers randomly drawn from the same pool.

Quiz B by gender). This constituted the "worker pool" for this experiment.

In a separate phase, the "main phase," I presented both hiring managers and clients (sample descriptions in Section 2.1.4) with 15 randomly selected pairs of workers drawn from the worker pool. For each pair, hiring managers and clients observed a "productivity signal" for each worker, consisting of their highest level of education and their score on the first of the two tasks (Quiz A). They also observed each worker's gender (see Figure 1, panel (b) and (c) for examples of these decision screens). Hiring managers and clients saw the same decision interface and the same worker pairs, but had different objectives. For each pair, clients were incentivized to select the most productive worker in the pair based on their score on the second task (*"select the worker with the highest Quiz B score"*). Meanwhile, hiring managers were incentivized to select the worker that a randomly chosen client selected (*"select the worker that the client chooses"*). Both clients and hiring managers received a bonus if they selected the correct worker in one randomly selected worker pair. To determine hiring manager bonuses, managers were yoked to a randomly selected client from their state of residence and received the bonus if they selected the same worker that the client selected in the randomly chosen pair. Hiring managers were aware of the information clients received and clients' incentives (e.g., *"A randomly selected 'Client' from the previous phase saw the same information about these workers."*). Thus, much like in many labor markets, hiring managers had an interest in hiring workers that they believed clients would endorse. Both hiring managers and clients had to correctly answer a set of comprehension checks to ensure they understood the instructions before being allowed to proceed to the task. Hiring managers and clients completed the same hiring task at both baseline (before the election) and endline (after the election).

2.1.2 Belief Elicitation

For hiring managers only, I measured beliefs about clients' average discrimination against female workers. To incentivize the accurate reporting of beliefs, I followed the procedure in Danz, Vesterlund, and Wilson (2022) and asked managers to provide their best estimates of the average proportion of female workers that clients would select, providing them with a bonus if they guessed within 3.3 percentage points of the true value.[4] This constituted my primary preregistered measure

[4] If this question was randomly selected for payout, they received an additional \$1.00 bonus. To check this measure, I asked them a 5-point qualitative Likert scale about how much on average clients favored male or female workers.

of third-party discrimination beliefs. I also measured both client and hiring manager beliefs about male and female worker productivity using the same procedure and incentives.[5] To measure hiring managers' more general beliefs about others' discrimination, not just those of clients in this setting, I also asked them to assess how many people out of 100 they believed supported female labor force participation (taken from the Gallup World Poll) and how supportive they believed others were (on a 7-point Likert scale) of jobs being reserved for men when they are scarce (reverse coded; taken from the World Values Survey). I collected these belief measures both before and after the election.[6] See Appendix Supplement A.2 for all belief questions, and OSF for all materials.

Additionally, to determine the ex post "surprisingness" of the election result for hiring managers, I measured their baseline beliefs about the election outcome in their state and at the national level. For each hiring manager, in the baseline survey I recorded which candidate they believed would win, the degree of certainty in their choice (on a 7-point Likert scale), and the margin by which they believed the candidate would win (in percentage difference in vote/electoral seat share between the first- and second-placed candidate). I incentivized all estimates of the election winner and the winning margin.[7] See Appendix Table C6 for a summary of these baseline beliefs.

2.1.3 Dependent Variables

My primary preregistered dependent variables include *changes* in hiring managers' beliefs about client discrimination and *changes* in the decision weight hiring managers placed on worker gender in their hiring choices. For beliefs, I calculated the difference in managers' incentivized beliefs about the proportion of female workers clients selected between baseline and endline surveys. I estimated the decision weight on worker gender separately at baseline and endline by regressing each manager's choices in the hiring task on worker gender and qualifications (i.e., workers' highest

These measures were highly correlated: $r = 0.38$, $p < 0.001$.

[5] I used a different threshold to determine the bonus to adjust for scale differences. Those within 5 percentage points of the correct average received a $1.00 bonus if this question was randomly selected for payout.

[6] The one exception is that I only collected hiring manager beliefs about male and female productivity in the endline survey. I did this to limit the number of beliefs elicited during the baseline survey to avoid fatigue during responding (as hiring managers also provided a battery of beliefs about the election at baseline), which might have affected the quality of other belief measures that were first order to identification.

[7] Participants were paid a bonus if they guessed the correct winner if that question was randomly selected for payout. For the winning margin, I used the same method as for the beliefs about worker productivity and client discrimination, providing a bonus if they estimated within 3 percentage points or 10 electoral seats of the actual winning margin for the state and national levels respectively, following the incentivization procedure in Danz et al. (2022).

level of education and Quiz A scores). This captures hiring managers' gender discrimination holding worker qualifications fixed. I then take the difference between the endline and baseline weight estimates. See the Analysis Supplement A.4 in the Appendix for more details on how these weights were derived for each manager. For ease of interpretation, I also report the proportion of female workers hired, which is highly correlated with the estimated gender weight ($r = 0.89$).

2.1.4 Sample

As preregistered, I recruited 600 participants on Prolific Academic (500 hiring managers and 100 clients), aiming for a final sample of at least 480 participants (400 hiring managers and 80 clients) completing both baseline and endline surveys after exclusions. I restricted recruitment to participants who resided in 11 U.S. states where the election outcome was uncertain prior to the 2024 presidential election, according to poll data. This included all 7 "swing states" (where the predicted margin of victory was less than 2 percentage points), and 4 other states where the predicted margin of victory was still relatively low (see the Appendix Sample Selection supplement for details about state selection and polling data). As preregistered, I screened out participants who failed attention checks, were living in a state outside of the specified set at the time of the surveys, or failed to complete the endline survey. After exclusions, the final sample included 497 participants (412 hiring managers and 85 clients) for the primary analyses (see Table A2 in the Appendix for details about this sample). Given that my primary analysis relies on difference-in-differences for only those that completed both baseline and endline surveys, attrition should not affect the internal validity of my findings. However, I find no evidence that the final sample differed from the sample recruited at baseline (see Tables A3 and A4 in the Appendix).

2.1.5 The Election as a Belief Shock

Elections have been shown to provide public signals of what others in society think or prefer, causing people to update their beliefs about third-party behavior. For instance, following Donald Trump's victory in the 2016 U.S. presidential election, perceptions of his local electoral success directly impacted public support for anti-immigration policies (Bursztyn, Egorov, & Fiorin, 2020). I propose that the 2024 U.S. presidential election could have caused similar shifts in beliefs about third-party gender discrimination, given the clear gendered valence of the candidates and parties involved. Prior to the election, Donald Trump had been described as "sexist," "misogynistic," and representative of a culture of "hypermasculinity" in the popular press, and was the target of several allegations of sexual assault (Bracic, Israel-Trummel, & Shortle, 2019; Filipovic, 2017; The Guardian, 2024a, 2024b). The global Women's March in 2017 was widely understood as a direct protest of his 2016 election (Bracic et al., 2019; Larreboure & Gonzalez, 2021), and his support has been linked to regressive views of women and increased discrimination against other marginalized groups more broadly.[8] By contrast, Kamala Harris would have been the first female U.S. president and was considered largely progressive regarding women's role in society (Armour, Appleby, & Rovner, 2024; NPR, 2024)—a divide reflected in the parties themselves: as of the election, 41% of Democratic representatives in Congress were women, compared with just 16% of Republican representatives (Pew Research Center, 2023), and the popular press extensively covered the gender gap in voting patterns separating the two parties (Page, Hoff, & Koch, 2024; The Economist, 2024; Zhou, 2024).

Taken together, Donald Trump winning the election could have provided hiring managers with information about others' preferences and beliefs regarding gender. As Prof. Laura Kray described in November 2024, "The results [President Trump's victory in the 2024 election] appear to confirm that most American voters still implicitly endorse traditional gender hierarchies..." (Lempinen, Pohl, & Thulin, 2024). Given the above, I argue that learning about a Donald Trump victory could have caused people to update their beliefs about third-party gender discrimination, and consequently affected their willingness to hire a female worker. To reduce noise in participants' awareness of the election outcome, the endline survey reminded hiring managers which candidate won the highest vote share in their state prior to eliciting their beliefs and hiring choices. I verify the proposed shifts in third-party beliefs in Section 2.2.1.

2.1.6 Identification Strategy Overview

My primary analysis tests how beliefs about third-party discrimination and the choice to hire female workers changed after the 2024 U.S. election as a function of (i) which candidate won

[8] Support for Trump in the 2016 election predicted regressive views of women in society (Setzler & Yanus, 2018) and related work documents increases in racial discrimination and hate crimes following his public statements and election victories (Edwards & Rushin, 2018; Feinberg et al., 2022; Grosjean et al., 2023; Müller & Schwarz, 2023).

in a hiring manager's state, and (ii) how unexpected this result was for each hiring manager (see Supplement A.4 for details). Several features of the design support identification. First, the analysis uses a difference-in-differences approach, where the primary dependent variables are changes from before to after the election: the change in manager beliefs about client discrimination, and the change in managers' likelihood of hiring a female worker. This removes the effect of time-invariant differences between hiring managers. By keeping the period between baseline and endline measurement short (12-17 days), I limited the scope for violations of the parallel trends assumption: it is unlikely that beliefs or hiring choices changed over this brief window for reasons unrelated to the election. For robustness, I run specifications including different sets of control variables to account for any deviations in the parallel trends assumption predicted by baseline characteristics, and test for changes in other outcomes as a placebo.

Second, I test for the effect of the election (as a form of belief shock about third-party discrimination) by both leveraging state-level variation in which candidate won the election, and by leveraging individual-level variation in how surprising this result was relative to hiring managers' baseline beliefs. To operationalize "surprisingness," I use deviations between the observed election outcome and participants' ex-ante predictions about the outcome. I propose that deviations should cause belief-updating about third-party discrimination (Bursztyn, Egorov, & Fiorin, 2020). I specifically focused on states where poll data suggested the election would be closely contested in order to increase variation in the surprisingness of the result (Jennings & Wlezien, 2018; Kotak & Moore, 2022): in states where the election was closely contested, participants should have been both less certain about which candidate would win, and more likely to wrongly predict the winning candidate (and thus be surprised by which candidate won the election). I leverage both categorical deviations from managers' predictions (i.e., failures to predict *which* candidate would win the election), and continuous deviations in terms of the winning candidates' vote share. I specifically prioritize the categorical surprise measure because decision makers tend to place more weight on outcomes that cross reference points or boundaries (Esponda et al., 2023; Naborn & Bogard, 2025; O'Donoghue & Sprenger, 2018). For example, a manager who expected Harris to win with 51% but saw her lose with 48% (a boundary-crossing deviation) might update their beliefs more than a manager who saw her win with 65% instead of an expected 68%, even though the magnitude of surprise (3 percentage points) is identical in both cases. For each analysis presented below, I describe which independent variable is used.

2.2 Results

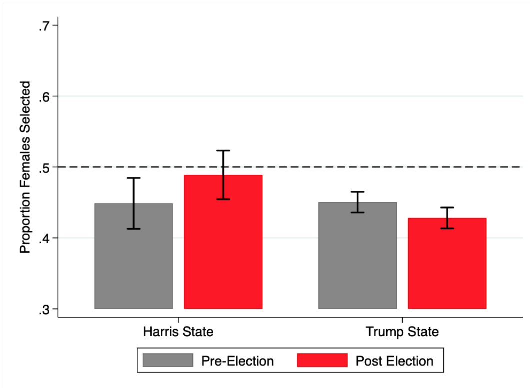
2.2.1 The Election Shifted Beliefs about Third-Party Discrimination

First, I test whether the election outcome affected beliefs about third-party discrimination. Figure 2, Panel (a) shows hiring managers' beliefs about the proportion of female workers that clients (in their same state) selected in the hiring task, split by pre- and post-election and by which candidate eventually won their state. Before the election, managers in states where Trump eventually won (Trump states) and states where Harris eventually won (Harris states) held similar beliefs about third-party discrimination: both groups expected clients to choose fewer female than male workers (selecting female workers approximately 45% of the time, significantly less than parity in a one-sample t-test: $\Delta = -0.050$, $t = -7.23$, $p < 0.001$). After the election, these beliefs diverged. Managers in Trump states expected clients to discriminate against female workers more after the election than before—the predicted rate of female worker selection dropped from 45.0% to 42.8% ($t = -2.53$, $p = 0.012$). This change was significantly different from that of managers in Harris states (Difference-in-difference=6.2 percentage points, $t = 2.83$, $p = 0.005$), whose expectations of clients' female worker selection, if anything, slightly increased (from 44.9% to 48.9%; $t = 1.98$, $p = 0.053$).[9] Regression analyses confirm this result and show it is robust to the inclusion of several sets of control variables (Appendix Table C7). Managers' beliefs about client gender discrimination therefore responded differently depending on the election outcome in their state.

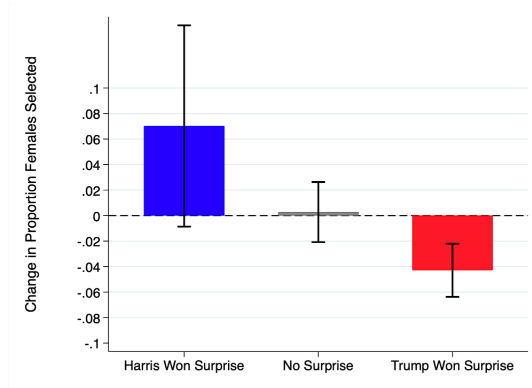
Next, I investigate how these belief changes varied as a function of managers' baseline beliefs about who would win the election—and thus, how surprising the election result was for each manager. Figure 2, Panel (b) shows the change in managers' beliefs about clients' female worker selection over the election period, split by whether managers were "surprised" by a Trump victory (i.e., they predicted Harris would win in their state, but Trump won), surprised by a Harris victory, or correctly predicted the election outcome. Changes in beliefs about clients' gender discrimination were largest for managers whose baseline beliefs about the election winner were incorrect. Managers

[9] Because Trump ultimately won 9 of the 11 closely contested states in the sample, only 66 of 412 hiring managers ultimately resided in Harris states, limiting statistical power to detect within-group changes for Harris-state managers.

Figure 2: Hiring Manager Beliefs about Clients' Choices: Proportion of Female Workers Clients Selected



(a) Pre-post by State Election Result



(b) Difference by "Belief Shock"

Notes: Panel (a) shows the mean hiring manager belief about the proportion of female workers clients selected, separately for (i) U.S. states in which Trump won the highest share of votes in the 2024 National Election and states in which Harris won the highest share, and (ii) pre-election (baseline) and post-election (endline) measures. The dotted line represents 50% male and female workers hired. Panel (b) shows the mean values for the change in these beliefs from baseline to endline (the change in the proportion of female workers that hiring managers believed clients selected). Positive values indicate that hiring managers' estimates of the number of women that clients selected increased after the election (relative to before), and negative values indicate these estimates decreased. Zero reflects no change in managers' beliefs about the number of women that clients selected. This is displayed separately for hiring managers that (i) failed to predict a Harris victory in their state (i.e., Harris won, but they predicted Trump would win), (ii) correctly predicted the winning candidate, (iii) failed to predict a Trump victory in their state. Error bars represent 95% confidence intervals of the means.

surprised by a Trump victory reduced their beliefs about the proportion of female workers that clients selected by 4.3 percentage points ($t = 4.03$, $p < 0.001$). Managers surprised by a Harris victory expected clients to hire 7.0 percentage points *more* female workers after the election than before, though this was not significant at $\alpha = 0.05$ ($t = 1.75$, $p = 0.093$). There is no evidence that managers who correctly predicted the election outcome changed their beliefs ($\Delta = 0.3\%$, $t = 0.23$, $p = 0.82$). Preregistered regressions confirm that this categorical surprise about the election winner (i.e., failing to anticipate *who* would win the election in their state) significantly affected managers' beliefs about clients' gender discrimination, including after controlling for which candidate actually won in each state (Appendix Table C7; column 5 includes state fixed effects).

A second preregistered measure of election surprise—the difference between managers' predicted and actual vote shares for the winning candidate in their state (a continuous measure of surprise)—also significantly predicted changes in beliefs about clients' gender discrimination (Appendix Table C7). Taken together, managers who failed to correctly predict the election outcome, and for whom the election was thus more informative, changed their beliefs about client discrimination more than those who predicted it accurately.

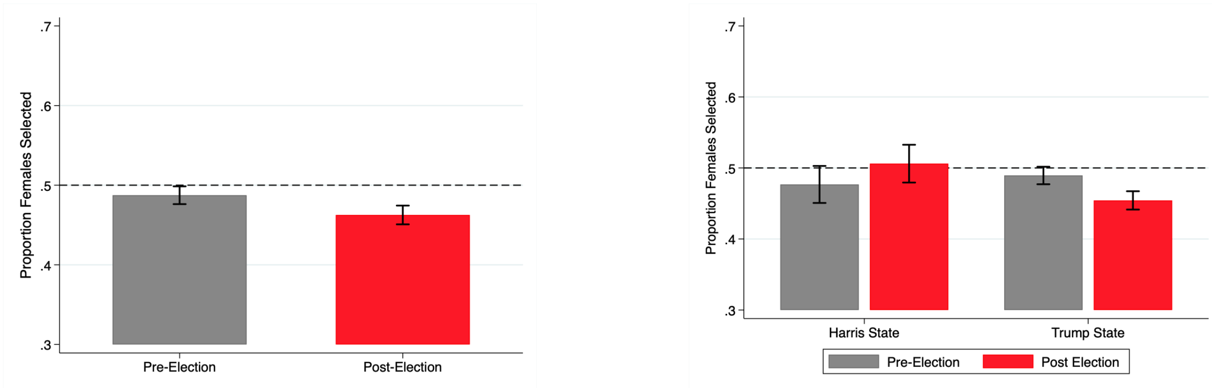
2.2.2 The Effect on Hiring Discrimination

The Effect of the Election on Hiring

The above subsection shows that hiring managers in Trump states expected clients to discriminate more against female workers after the election than before, a belief change which did not occur for managers in Harris states. How did managers' hiring decisions correspond to these belief changes? First, Figure 3, Panel (a) shows that, on average, hiring managers across all surveyed states hired fewer female workers after the election than before ($\Delta = 2.48\%$, $t = 3.53$, $p < 0.001$). Panel (b) splits this result by which candidate won the election in each manager's state. As with beliefs, the overall reduction in female worker hiring was driven by managers in Trump states, who hired 3.5% fewer female workers after the election than before ($t = 4.61$, $p < 0.001$), while managers in Harris states hired 2.9% more female workers (though this was not significant at $\alpha = 0.05$; $t = 1.76$, $p = 0.082$). Figure 4, Panel (a) shows the same pattern of results for my primary dependent variable: the change in the weight managers placed on worker gender in their hiring choices, holding worker qualifications fixed (see Section 2.1.3 for a description of this variable). Managers in Trump states were less likely to hire female workers (relative to male workers) after the election than before ($p < 0.001$), holding qualifications constant, while this likelihood increased for managers in Harris states ($p = 0.017$): this difference in weight change between Trump and Harris states was highly significant ($\beta = -0.071$, $t = -4.11$, $p < 0.001$; see Table 1 for the regression results).

The surprisingness of the election result, given managers' baseline expectations about which candidate would win, also affected the decision of whether to hire a female worker. Figure 4, Panel (b) shows that managers who expected Harris to win in a state where Trump eventually won were significantly less likely to hire a female worker after the election than before, while managers who failed to predict a Harris victory hired more female workers after the election. The preregistered regression confirms that this failure to anticipate the election winner significantly predicted the change in a manager's likelihood of hiring a female worker ($\beta = -0.035$, $t = -2.95$, $p = 0.003$; Table 1, column 1). This effect becomes marginally significant when I include state fixed effects ($t = -1.90$, $p = 0.059$); this is largely because managers who correctly anticipated Trump's victory

Figure 3: Hiring Managers' Hiring Choices: Proportion of Female Workers Selected

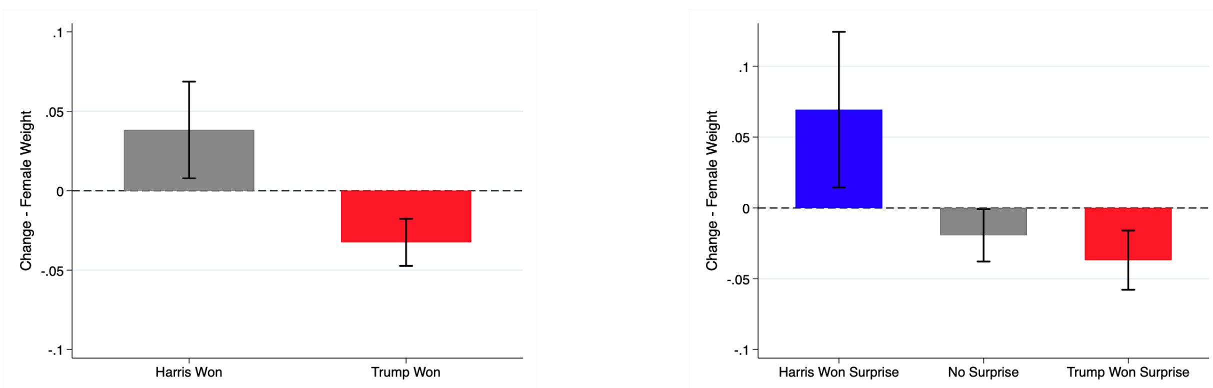


(a) Pre and Post Election

(b) Pre-post by State Election Result

Notes: Panel a) shows the mean proportion of female workers hired by hiring managers before and after the election, across all hiring managers. Panel (b) shows the mean proportion of female workers hired by hiring managers, separately for i) U.S. states in which Trump won the highest share of votes in the 2024 National Election and states in which Harris won the highest share, and ii) pre-election (baseline) and post-election (endline) measures. The dotted line represents 50% male and female workers hired. Error bars represent 95% confidence intervals of the means.

Figure 4: Hiring Managers' Hiring Choices: Decision Weights on Gender



(a) Difference by State Election Result

(b) Difference by "Belief Shock"

Notes: Panel (a) shows the mean change in the decision weight that hiring managers placed on gender over the election (positive values indicate weighing female gender more positively after the election than before, and negative weights indicate weighing female gender more negatively after the election). These weight changes represent the change in the likelihood of hiring female workers vis-a-vis male workers, holding worker qualifications constant. Panel (b) again shows the mean values for this change in decision weight, but separately for hiring managers that i) failed to predict a Harris victory in their state (i.e., Harris won, but they predicted Trump would win), ii) correctly predicted the winning candidate, iii) failed to predict a Trump victory in their state. Error bars represent 95% confidence intervals of the means.

in their state *also* reduced female worker hiring, shrinking the within-state difference between surprised and unsurprised managers (see Table C11). The secondary, continuous measure of election surprise—the difference between the actual and expected margin of victory in the manager's state—did not significantly predict changes in female worker hiring in the main regression, though results were significant in specifications that account for outliers in election beliefs (Table 1; Appendix

Figure A1).[10]

Table 1: Regression Results for Primary Outcome,
Change in Gender Weight in Hiring, by "Belief Shock"

	1	2	3	4	5	6	7	8
Trump won state	-0.071***	-0.068***	-0.067***	-0.068***		-0.063***	-0.064***	-0.06***
	(0.017)	(0.017)	(0.017)	(0.017)		(0.018)	(0.015)	(0.015)
Surprise: election winner	-0.035***	-0.034***	-0.034***	-0.032**	-0.03*	-0.029**	-0.027***	-0.026***
	(0.012)	(0.012)	(0.012)	(0.013)	(0.016)	(0.013)	(0.01)	(0.01)
Surprise: winning margin	-0.031	-0.035	-0.034	-0.026	-0.03	-0.025	-0.02	-0.021
	(0.021)	(0.022)	(0.022)	(0.024)	(0.023)	(0.024)	(0.018)	(0.019)
	Main	Control	Control + Productivity belief	Control + Politics	Control + State Fixed Effects	All controls	Baseline Belief	Baseline Belief + Controls

Notes: Each column and row correspond to a separate OLS regression specification, with coefficients on the primary independent variable reported and robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Each row represents regressions with a different set of independent variables (different operationalizations of the "belief shock"), representing the β_1 coefficient in regression equations 1, 2, and 3 respectively (shown in the Supplement A.4). The dependent variable (in columns 1-6) is the change in the weight the hiring manager places on gender (the weight placed on the worker being female) when making hiring choices from before to after the election, calculated by subtracting the baseline from the endline gender weights, determined by hiring managers' choice data. Control variables vary by specification as indicated, but the base controls (columns 2-6 and 8) include dummy variables for black and white race, male gender, college educated (or not), and continuous age and age squared. Productivity belief indicates incentivized beliefs about the relative male and female worker productivity, measured by predicted task scores; Politics refers to political party affiliation fixed effects, and "All Controls" represent all the above control variables, and additionally income and highest education level fixed effects. Columns 7 and 8 provide results for regressions using the alternative specification recommended by McKenzie (2012), which regresses endline gender weight on my key independent variables, controlling for baseline gender weight (rather than using the change in gender weight as the dependent variable).

The Effect of Third-Party Beliefs on Hiring

I have demonstrated that the 2024 U.S. presidential election result and its surprisingness produced both changes in beliefs about third-party (client) discrimination and changes in hiring choices. To test the relationship between these belief changes and hiring, I use instrumental variable (IV) regression analysis as per my preregistration, using the same set of independent variables—the variation in which candidate won the election in each state, and in the election "surprisingness"—as instruments for the change in beliefs about client discrimination. IV regression requires that the instruments both predict changes in third-party discrimination beliefs (the "first-stage") and do not affect hiring through any channel other than these belief changes (the "exclusion restriction"). Section 2.2.1 shows that the election affected managers' beliefs about clients' discrimination, fulfilling the first stage. The exclusion restriction cannot be verified directly; however, in the robustness sub-section below I present evidence against the most plausible alternative explanations for the election's effect on hiring—including changes in beliefs about female worker productivity, the em-

[10] This analysis is sensitive to outliers in managers' predictions about the margin of victory. When removing outliers, or when using an alternative measure of certainty (a Likert scale of confidence in their predicted winner), the coefficients on this marginal surprise measure are significant (see Appendix A.5).

boldening of would-be discriminators, and broader changes in decision-making weights. In the IV regressions, I find that managers' beliefs about clients' discrimination predict managers' own discrimination (Table 2). Together, these results indicate that beliefs about third-party discrimination were an important channel through which the election affected hiring choices.

Table 2: Instrumental Variable Regression Results:
The Effect of Belief Changes on Hiring

	1	2	3	4	5	6	7	8
Trump won state	1.131**	1.149**	1.141**	1.203**		1.072**	1.01**	1.01**
	(0.488)	(0.529)	(0.524)	(0.565)		(0.509)	(0.417)	(0.443)
Surprise: election winner	0.681**	0.689**	0.69**	0.704*	0.601	0.635*	0.532**	0.527**
	(0.288)	(0.299)	(0.299)	(0.359)	(0.382)	(0.345)	(0.235)	(0.243)
Surprise: winning margin	0.471	0.506	0.496	0.424	0.436	0.423	0.302	0.305
	(0.365)	(0.375)	(0.37)	(0.442)	(0.377)	(0.442)	(0.289)	(0.294)
	Main	Control	Control + Productivity belief	Control + Politics	Control + State Fixed Effects	All controls	Baseline Belief	Baseline Belief + Controls

Notes: Each column and row correspond to a separate two-stage-least-squares (2SLS) regression specification. The primary statistic is the coefficient on the change in hiring managers' beliefs about client discrimination (the primary independent variable, reported in the second-stage of the 2SLS). Thus, each coefficient represents an estimate of the causal effect of changes in beliefs about client discrimination on the likelihood of hiring a female worker. Standard errors for each coefficient in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Each row represents regressions with a different set of instruments for the change in beliefs about client discrimination, corresponding to the variables $TrumpState_i$, $ElectionSurprise_i$, and $ElectionSurpriseDelta_i$ from regression equations 1, 2, and 3 respectively as instruments (shown in the Supplement A.4). The dependent variable (Y_i in columns 1-6) is the change in the weight the hiring manager places on gender (the weight placed on the worker being female) when making hiring choices from before to after the election, calculated by subtracting the baseline from the endline gender weights, determined by hiring managers' choice data. Control variables vary by specification as indicated, but the base controls (columns 2-6 and 8) include dummy variables for black and white race, male gender, college educated (or not), and continuous age and age squared. Productivity belief indicates incentivized beliefs about the relative male and female worker productivity, measured by predicted task scores; Politics refers to political party affiliation fixed effects, and "All Controls" represent all the above control variables, and additionally income and highest education level fixed effects. Columns 7 and 8 provide results for 2SLS regressions using the endline gender weight as the dependent variable, using my key independent variables as instruments for the change in beliefs about client discrimination, controlling for baseline gender weight (rather using the change in gender weight as the dependent variable).

Robustness of the Hiring Effects

The effects of the election on female worker hiring—specifically, of which candidate won in a state and whether the winning candidate was predicted—are robust to different specifications and to tests of alternative explanations. First, these results are highly significant ($p < 0.004$ in the main specifications; Table 1) and relatively consistent across different types of hiring managers, including both Republican and Democrat managers (Appendix Figure B6). Second, by leveraging within-manager differences between baseline and endline surveys, I removed the effects of time-invariant differences between managers across states and surprise categories. Third, the very short period between surveys limited potential violations of parallel trends: any confounding factor would need to have affected female relative to male worker hiring in the 12-17 days between surveys, and to have done so differently across Trump vs. Harris states and across surprise categories. Finally, managers across Trump and Harris states and across surprise categories did not differ at baseline in their third-party beliefs or hiring choices (Figures 2, 3, B3, and B2), and effects were robust to the inclusion of different sets of control variables (Table 1).[11] Together, these factors suggest that selection, violations of parallel trends, or mechanical artifacts (such as ceiling effects or regression to the mean) are unlikely to account for the observed changes in hiring over the election.

Supporting the proposition that the election affected hiring through third-party beliefs rather than through other channels, beliefs and decision patterns unrelated to third-party discrimination did *not* change over the election. First, clients' own hiring choices did not change over the election period (Appendix Figures B11 and B12).[12] Clients were drawn from the same population and experienced the same election as hiring managers; if the effects on managers' choices resulted from the election increasing the salience of gender, emboldening decision makers with pre-existing gender prejudice, or some other general shift, clients' choices should also have moved against female workers. That they did not suggests the election affected choices through beliefs about *others'* discrimination—a channel absent for clients. Second, beliefs about the relative productivity of male and female workers did not change over the election, did not predict changes in female worker hiring, and did not meaningfully alter the main effect estimates when added to regressions (Figure B13; Table 1).[13] Third, while the weight managers placed on worker gender changed over the election, there were no such changes in the weights placed on workers' education and Quiz A scores (Appendix Figure B7), suggesting the election did not broadly change decision-making patterns. Reflecting the above, the IV regression estimates of the effect of belief changes on hiring are largely robust to different specifications (Table 2). Taken together, the effects of the election on female worker hiring are consistent with the third-party belief mechanism. Nevertheless, a natural experiment cannot definitively rule out all alternative channels. The controlled experiment (Section 3) addresses this limitation directly, exogenously manipulating beliefs about third-party discrimination to isolate

[11] The effect of a Trump relative to Harris victory remains between 7.1 and 6 percentage points and significant in all specifications. The effect of the categorical "surprise" variable remains between 2.6 and 3.5 percentage points and is significant in all specifications but one, where $p = 0.059$.

[12] The client sample is smaller than the hiring manager sample, limiting the precision of these estimates. However, the sample is sufficient to rule out negative effects on female worker selection of even half the magnitude observed for hiring managers.

[13] Belief changes were estimated using client beliefs about male and female worker productivity, as I only collected hiring managers' endline beliefs. These manager endline beliefs did not predict hiring changes nor affect estimates of the election's effect in regressions.

their causal effect on hiring choices.

2.2.3 Are Hiring Managers Accurate?

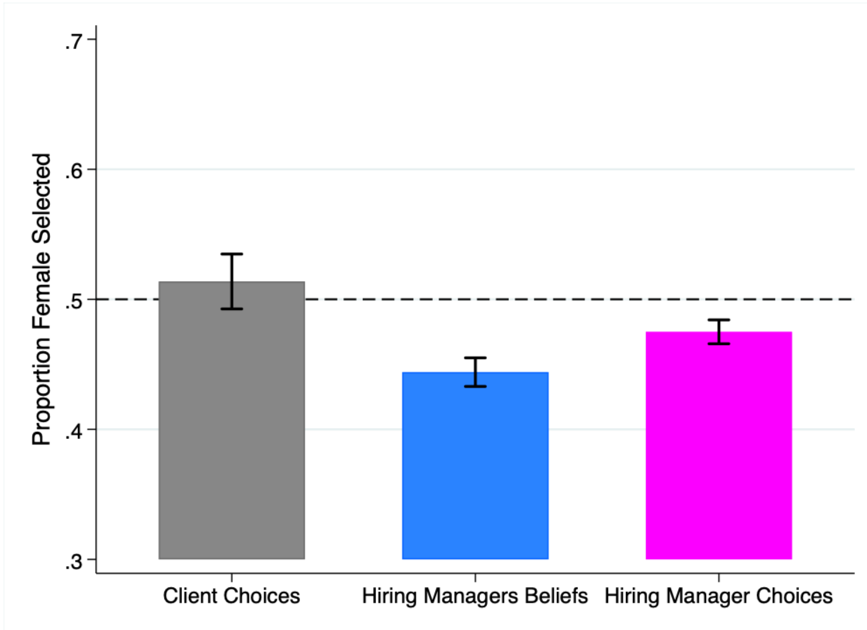
Hiring managers thus changed their hiring decisions based on their beliefs about client discrimination; but were these beliefs accurate? To test this, I compare managers' beliefs about clients' female worker selection to clients' actual choices, pooling across baseline and endline surveys as preregistered. Hiring managers significantly overestimated clients' discrimination against female workers: managers believed clients would choose female workers 44.4% of the time, 6.9 percentage points less than clients' actual female worker selection (Figure 5; one sample t-test of hiring manager beliefs equaling clients' mean selection: $t = 5.25$, $CI = [0.044; 0.096]$, $p < 0.001$). To test whether this overestimation led hiring managers to exacerbate discrimination in this labor market, I pooled decision data for clients and hiring managers and regressed gender decision weights (the weight they placed on a worker being female) on a dummy variable for whether the decision maker was a hiring manager. Hiring managers were 6.1 percentage points less likely than clients to hire an equivalent female worker (Appendix Figure B8; $t = 4.92$, $CI = [-0.087; -0.037]$, $p < 0.001$). This effect is robust to the inclusion of different sets of control variables (Figure B8), and similar for different sub-samples of hiring managers: for all 10 tested sub-samples, point estimates are within the 95% confidence interval of the main effect and statistically less than 0 (Appendix Figure B10).[14] Taken together, hiring managers discriminated against female workers more than clients did, and more than they would have with accurate beliefs about clients' choices.

3 Controlled Experiment

To more cleanly identify the causal effect of third-party discrimination beliefs on hiring choices, I conducted a preregistered online experiment where I directly manipulated manager beliefs about client gender discrimination, rather than relying on naturally occurring variation in these beliefs (AsPredicted #179434). I conducted this experiment using the same hiring paradigm as in the

[14]Note that Figure 5 also shows that managers' choices did not fully reflect their beliefs about client discrimination: they hired female workers 47.5% of the time while believing clients selected them 44.4% of the time ($t = 5.16$, $p < 0.001$). This deviation—which could reflect differences between choice and belief-elicitation methods, or a reluctance to fully act on discriminatory beliefs—implies that managers' inaccurate beliefs translated into less additional discrimination than full belief-following would have produced.

Figure 5: Comparing Hiring Manager Beliefs and Choices to Client Choices



Notes: The gray (left-most) bar represents the mean proportion of female workers clients selected. The blue (middle) bar represents hiring managers' mean belief about the proportion of female workers clients selected. The magenta (right-most) bar represents the mean proportion of female workers hiring managers hired. In all cases, these means are for clients and hiring managers across all states and collapsed across baseline and endline choices and beliefs (as-preregistered). Error bars represent 95% confidence intervals of the means

election experiment, but included a set of treatment conditions where I varied the information provided to hiring managers and clients. Below I present the details of this design.

3.1 Design

I recruited participants from Prolific Academic who again played one of three roles in a stylized online labor market: workers, clients, or hiring managers. I used the same pool of workers as in the election experiment (Section 2.1.1) but recruited a new set of clients and hiring managers, with no restriction on U.S. state of residence. The final sample included 502 participants: 300 hiring managers and 202 clients. The design mirrored the election experiment: hiring managers and clients completed the same hiring task, choosing between workers in each of 15 randomly assigned worker pairs where they observed each worker's gender, highest level of education, and score on an online task (see Section 2.1.1 for more details about this task). Clients were incentivized to select the most productive worker in each pair and hiring managers were incentivized to select the worker that a randomly yoked client selected. Hiring managers again provided incentivized beliefs about client discrimination and about male and female worker productivity. Unlike in the election experiment, participants completed the hiring task and belief measures only once, outside the election period (please find materials for the experiment here on OSF).

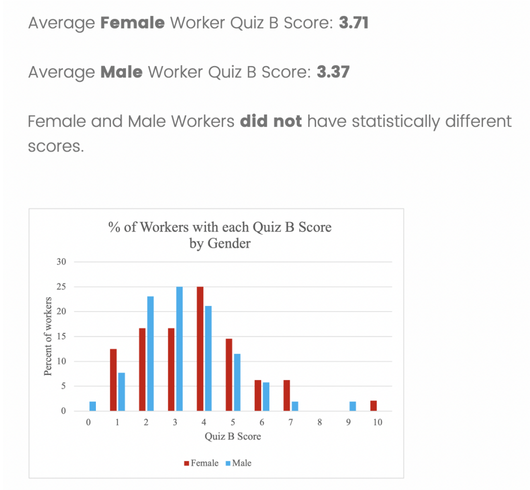
3.1.1 Treatment Conditions

Before completing the hiring task, I experimentally manipulated hiring managers' beliefs about clients' gender discrimination by randomly assigning managers to one of three experimental conditions. In the *Control* condition, managers completed the hiring task with no information about the true association between worker gender and productivity (i.e., Quiz B scores). Meanwhile, in the two treatment conditions, I showed managers the true distribution of worker productivity by gender, which revealed that female workers slightly outperformed male workers, though the difference was small and not statistically significant (Figure 6). Hiring managers assigned to the *Clients Debiased* condition were told that clients would see "exactly the same" information, while managers assigned to the *Clients as-is* condition were told that clients would "not see" this information. To ensure the design was incentive compatible and deception-free, clients were themselves randomly assigned to see this information or not, and each manager was yoked to a client from the corresponding condition (see Figure 6 for a diagram of all conditions and yokings). All condition-specific information was provided before participants completed the hiring task and belief measures.

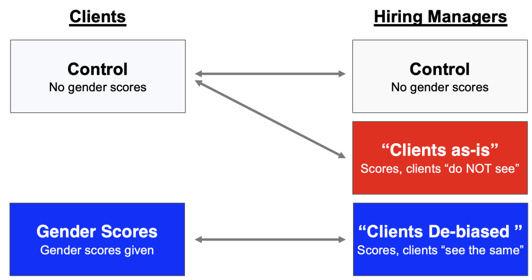
I predicted that hiring managers would expect clients to discriminate against female workers in the absence of worker productivity information, given the gender-stereotyped nature of the task and previous evidence of discrimination against female workers on this exact task (Bohren et al., 2022).[15] Managers who learned that clients saw the true relationship between productivity and gender should instead expect clients to be more favorable toward female workers. I verified this belief manipulation using both pilot data and data from this experiment (Section 3.2.1). Importantly, by providing managers in both treatments with the true distribution of worker scores by gender, I rule

[15] There is a history of male-dominance in math, finance, history, and related tasks (Bohren et al., 2022; Bordalo, Coffman, Gennaioli, & Shleifer, 2019; Coffman, 2014; Guiso et al., 2008); female workers are also prone to self-stereotype in similar tasks, thus believing that they themselves are worse in stereotyped domains than they actually are (Coffman, 2014). Finally, previous work using the exact same task found that Prolific workers in the U.S. a) expected female workers to perform worse than male workers, and consequently discriminated against them in a hiring task, and b) believed that others also discriminated against female workers based on hiring in this task (Bohren et al., 2022).

Figure 6: Controlled Experiment: Treatment Groups and Gender Score Information



(a) Information: Gender Score Distribution



(b) Treatment Conditions and Client Yoking

Notes: Panel (a) shows the "gender score information" provided to clients in the "Gender Score" condition and to hiring managers in the *Clients as-is* and *Clients Debiased* conditions in the Controlled Experiment. Panel (b) shows the treatment conditions for both clients and hiring managers in the Controlled Experiment and arrows represent which client treatment conditions hiring managers in each condition were randomly paired with to determine their bonuses.

out traditional statistical discrimination (i.e., differences in beliefs about productivity by worker gender) as an explanation for differences in female worker hiring between these groups; I verify that beliefs about worker productivity did not differ between them.

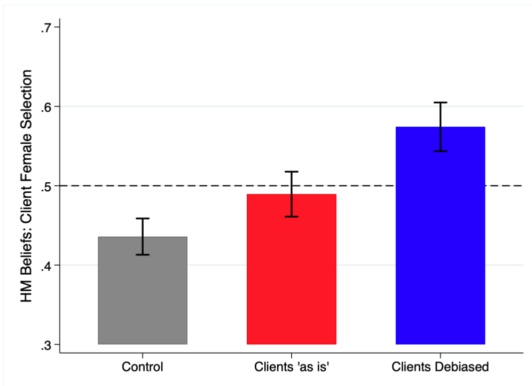
3.2 Results

3.2.1 Beliefs about Client Discrimination

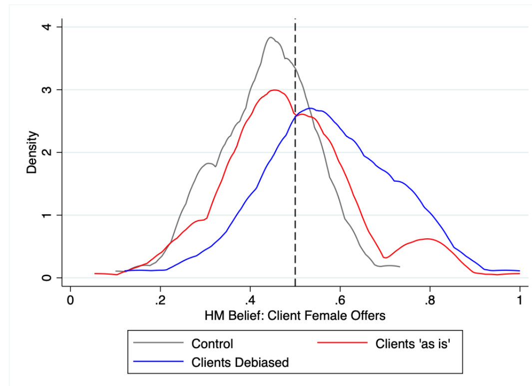
First, I test whether the belief manipulation worked, focusing on hiring managers' incentivized beliefs about the proportion of female workers clients selected. Learning that clients received productivity information broken down by worker gender changed managers' beliefs about client discrimination: managers in the *Clients Debiased* condition believed that clients selected more female workers than did managers in the *Clients as-is* condition ($\beta = 0.085$, $t = 3.98$, $CI = [0.043; 0.127]$, $p < 0.001$; Figure 7; Appendix Table F15). Surprisingly, receiving the gender-productivity information was itself sufficient to change beliefs: managers in the *Clients as-is* condition believed that clients hired more female workers than did managers in the *Control* condition ($\beta = 0.053$, $t = 2.78$,

$CI = [0.016; 0.091]$, $p = 0.006$).[16]

Figure 7: Controlled Experiment: Hiring Manager Beliefs about Client Female Worker Selection by Treatment Condition



(a) Means



(b) Kernel Density Functions

Notes: Panel (a) shows hiring managers' mean beliefs of the proportion of female workers selected by clients, in each treatment condition in the controlled experiment. The dotted line represents 50% male and female workers selected by clients. Error bars represent 95% confidence intervals of the mean belief. Panel (b) represents the kernel density function of these beliefs by treatment condition.

3.2.2 The Effect on Hiring Discrimination

Next, I examine the effect of these belief manipulations on hiring choices. Managers in the *Clients Debiased* condition hired significantly more female workers than managers in the *Clients as-is* condition ($\Delta = 0.061$, $t = 3.43$, $CI = [0.026; 0.096]$, $p < 0.001$; Figure 8). Because managers in both conditions received identical information about worker productivity by gender—differing only in what they were told clients saw—this difference isolates the effect of beliefs about *client* behavior, over and above any effect of the productivity information itself. In the primary pre-specified regression, which controls for workers' signals of productivity, the *Clients Debiased* treatment caused an 8.0 percentage point increase in the likelihood that managers hired a female worker ($t = 3.33$, $CI = [0.033; 0.127]$, $p = 0.001$; Appendix Table F15), an effect robust across specifications. As with beliefs, the gender score information alone also affected choices: managers receiving this information were 6.3 percentage points more likely to choose a female worker than those in the *Control* condition ($t = 2.96$, $CI = [0.021; 0.104]$, $p = 0.003$). Finally, a pre-specified instrumental

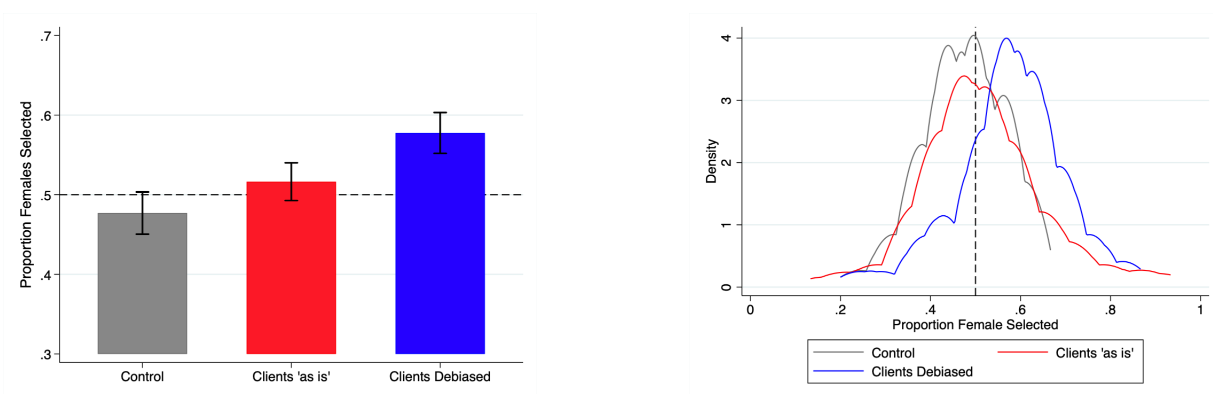
[16]This is surprising because managers in both groups were yoked to the same clients and had identical information about clients' choices. The finding is consistent with the curse of knowledge—decision makers who learn new information are unable to suppress it when making judgments (Camerer, Loewenstein, & Weber, 1989)—and with "information projection," where decision makers project their own knowledge onto others, even when others cannot possibly hold it (Madarász, 2012).

variable regression—using random assignment to the *Clients Debiased* treatment as an instrument for beliefs about client discrimination—shows that these third-party beliefs causally affected hiring choices (See Appendix Table F16).[17]

The effect of third-party discrimination beliefs on hiring also varied with workers' relative qualifications: manipulating these beliefs had the largest effect on female worker hiring when male and female workers in a given worker pair were more similarly qualified (see Appendix D.2).

Finally, I replicate the belief-accuracy results from the election experiment: hiring managers again significantly overestimated clients' discrimination against female workers (by 6.0 percentage points, $t = 5.11$, $p < 0.001$) and consequently discriminated more than clients did, hiring equivalent female workers 5.4 percentage points less often ($t = 2.91$, $p = 0.002$; see Appendix D.1 for full results).[18]

Figure 8: Controlled Experiment: Hiring Choices by Treatment Condition



(a) Means

(b) Kernel Density Functions

Notes: Panel (a) shows the mean proportion of female workers hired by hiring managers in each treatment condition in the controlled experiment, and the dotted line represents 50% male and female workers hired. Error bars represent 95% confidence intervals of the mean. Panel (b) represents the kernel density function of the proportion of female workers hired by treatment condition.

4 Discussion and Conclusion

This paper finds that hiring managers' beliefs about clients' discrimination causally affect their own hiring choices. First, Donald Trump, as opposed to Kamala Harris, winning the highest

[17]For the instrumental variable regression, the previous subsection establishes a strong first stage: the treatment affected beliefs about client discrimination. Random assignment, and the fact that the manipulation differed only in whether clients "did see" vs. "did not see" the gender-productivity information, support the exclusion restriction.

[18]As preregistered, these analyses use only participants assigned to the *Control* conditions (for both hiring managers and clients), whose beliefs were not manipulated, allowing a direct comparison to the election experiment where no productivity-by-gender information was provided.

vote share in a hiring manager's state in the 2024 U.S. presidential election increased managers' beliefs about gender discrimination by clients in their state, and in turn caused these managers to hire fewer female job candidates. Second, experimentally manipulating hiring managers' beliefs about client discrimination, holding other factors constant, significantly changed the share of female workers they hired. Across both studies, hiring managers overestimated client discrimination, and consequently hired fewer female workers than they would have given accurate beliefs. Given that clients did not discriminate against female workers on average, this led to a special form of self-fulfilling prophecy: hiring managers discriminated against female job candidates exclusively because they wrongly anticipated that clients would discriminate, when in fact they did not.

The present work thus builds on research discussing the importance of discrimination by customers and other third parties in perpetuating labor market discrimination. Gary Becker (1993) said in his 1993 Nobel lecture: "Of greater significance empirically is the long-run discrimination by employees and customers, who are far more important sources of market discrimination than employers." Following Becker, empirical work shows that the demographic characteristics of an organization's clients predict the demographics of the workers they hire (Bar & Zussman, 2017; Holzer & Ihlanfeldt, 1998; Neumark et al., 1996). Related research shows that the extent to which a position is "client-facing" also predicts gender and racial wage gaps and discrimination (Hurst et al., 2024; Kline et al., 2022). For instance, Kline et al. (2022) shows that both racial and gender discrimination were highest in sales and in retail positions with a high degree of customer interaction. However, this previous work has not experimentally tested whether beliefs about client discrimination drive these relationships, nor how these beliefs are formed. The present work directly measures and manipulates beliefs about client discrimination, testing their causal effect on hiring. The present work also adds to this existing literature by showing that public signals, such as elections, can affect these third-party beliefs.

The present paper thus links the research on customer discrimination to recent work in behavioral economics using controlled experiments in stylized labor markets to causally identify the effect of specific beliefs and belief-formation processes on discrimination (Barron et al., 2024; Bohren et al., 2023; Campos-Mercade & Mengel, 2023; Chang, Kirgios, Rai, & Milkman, 2020; Coffman et al., 2021; Esponda et al., 2023; Fischbacher, Kübler, & Stüber, 2024; Hagmann, Sajons, & Tinsley, 2026). This body of research has also documented the role of third-party beliefs in affecting discrimination—through a failure to account for discrimination by previous evaluators (Bohren, Hull, & Imas, 2025; Bohren, Imas, & Rosenberg, 2019) and through foreseeing discrimination by certain types of managers at later stages of the hiring process (Hirmas & Hausfeld, 2024). The present work builds on these by using a similar stylized labor market paradigm to document the influence of anticipated discrimination on final hiring decisions.

The finding in this paper that hiring managers overestimated client discrimination against female workers is consistent with related work showing that in most countries people systematically overestimate others' opposition to female labor force participation (Bursztyn et al., 2023). This comports with research on "pluralistic ignorance," showing that people assume the beliefs and actions of others differ from their own when they are in fact the same (Prentice & Miller, 1996). It is also consistent with related work on the "bias blind spot," showing that people tend to believe others are more biased than themselves (Pronin et al., 2004, 2002). To the extent that gender discrimination is perceived as a bias, these social misperceptions might help to explain the overestimates of discrimination I observe in the present study. Similar work shows that correcting these misperceptions can produce changes in female labor force participation. In an experiment in Saudi Arabia, Bursztyn, González, and Yanagizawa-Drott (2020) showed that correcting men's beliefs about others' support for female labor force participation (men systematically underestimated this support) increased their wives' likelihood of being employed. This is consistent with the findings in the controlled experiment in the present paper, where showing hiring managers that clients are "debiased," and thereby changing their beliefs about clients' gender discrimination, increased the proportion of female workers that they hired.

While I explore this phenomenon with hiring managers and clients, this work could have implications for other managerial settings and for a range of third-party beliefs. For example, organizational decision makers might anticipate discrimination by a range of different actors, e.g., a firm's operational or team managers, existing junior employees, its shareholders, or even its suppliers. Third-party beliefs might be especially relevant for certain types of recruitment, where third parties are first-order determinants of candidate success. For example, recruitment intermediaries' success relies on the final hiring managers' approval of their chosen candidate. Previous work shows that hiring intermediaries, such as external recruitment agencies, are less likely to include female candidates in their shortlists (see Fernandez-Mateo & King, 2011 and Fernandez-Mateo & Fernandez, 2016), partly because they anticipate gender bias from certain types of managers, such as male or White male managers (Hirmas & Hausfeld, 2024; Lide & Berg, 2023). Hiring for executive or board of director positions might be subject to similar pressures. The hired candidate's success in these roles relies heavily on the endorsement of shareholders, existing employees, or customers. This mechanism might help explain why women or racial minorities face particularly pronounced under-representation in these roles (Evans, 2022; Fernandez-Mateo & Fernandez, 2016; Gupta, Han, Mortal, Silveri, & Turban, 2018).[19]

This work also has implications beyond hiring contexts. For instance, home sellers might discriminate against Black home buyers if they anticipate well-documented racial discrimination in the credit market, as these buyers might be unable to finance their home (Hurtado & Sakong, 2023; Ladd, 1998; Pope & Sydnor, 2011). Likewise, political parties or voters might disfavor prospective female candidates because they anticipate voter discrimination against them (Corbett, Voelkel, Cooper, & Willer, 2022). Some evidence shows that voters strategically withheld votes from female candidates because they believed *others* wouldn't vote for them, making a vote for these candidates less impactful (Corbett et al., 2022). Additionally, Abraham (2020) found that entrepreneurs in male-typed occupations favored male over female contacts in a networking exchange, but only when the exchange involved a third party. It follows both theoretically and from these examples that when third-party discrimination affects decision maker outcomes, beliefs about third parties could have upstream effects on decisions, which substantially widens the scope for the effects of these beliefs across economic settings. Moreover, in cases where these beliefs are overestimated, existing discrimination could be exacerbated, as observed in the studies in this paper. This could occur, for example, in situations where individual attitudes change while collective recognition of this change in society lags (Bicchieri & Mercier, 2014; Bursztyn, González, & Yanagizawa-Drott, 2020). Or when others' discriminatory behavior is highly salient and psychologically "available", such as when there is extensive media reporting about discrimination (Pronin, 2007; Pronin et al.,

[19] Evans (2022) demonstrates that the under-representation of minority board members is largest when firms outsource board recruitment to a recruitment agency. This is consistent with the prediction that this might exacerbate the role of third-party beliefs in driving discrimination, as discussed in Fernandez-Mateo and Fernandez (2016).

2002).

Several limitations of this study merit discussion. First, I cannot fully rule out the concern that participants may have responded in socially desirable ways, although the study was designed to minimize demand effects and social desirability bias. For instance, all responses were fully anonymous, and all decisions and belief estimates were incentivized, with accuracy bonuses that were large relative to participants' base pay. Reflecting this, participants were highly responsive to worker qualifications in their choices, choosing the dominant worker in a pair 95.3% of the time, irrespective of worker gender (I provide details on this and the points below in the Appendix). Next, the hiring task was specifically constructed to reduce the salience of gender and to provide decision makers with "cover" for selecting either male or female workers in each decision. Given that the survey was already anonymous, this should have additionally reduced participants' reputational concerns about responding in certain ways. Finally, while social desirability might pose a concern for the levels of female worker hiring in the studies, it should not necessarily drive the differences in hiring observed between treatment conditions.

Next, although the experiments in this paper were incentive compatible and involved real hiring processes, a key limitation is that these findings were drawn from online samples in a stylized task rather than in the field. While these experiments were internally valid and leveraged a real-world belief-shock, it is difficult to generalize the effect sizes observed here to what one might expect for hiring outcomes in a field setting. Similarly, these studies cannot capture potential general equilibrium effects that might exist in real-world labor markets. For instance, although I find that hiring managers discriminated against female workers due to inaccurate beliefs about client discrimination, managers may update beliefs in practice through feedback from observed outcomes—such as client behavior, peer feedback, or firm performance—leading to corrections in these beliefs over time. However, in organizational settings, such feedback is often noisy, difficult to acquire, and challenging for managers to integrate, impeding learning (Esponda, Vespa, & Yuksel, 2024; Feng & Seasholes, 2005; Fiol & Lyles, 1985; Levitt & March, 1988; Niederle & Vespa, 2023). Moreover, if managers avoid hiring women because they anticipate client discrimination, they may rarely observe client reactions to female workers. This endogenous feedback could prevent learning even when signals would otherwise be informative (Denrell, 2005; Larrick & Wu, 2007). It is thus possible that these belief distortions persist in equilibrium (Denrell, 2005; Esponda, 2008; Niederle & Vespa, 2023). In fact, I find that managers with prior hiring experience are not more accurate in their beliefs about client discrimination than those with no experience (see Figure B9).

An important direction for future work is thus to examine how beliefs about third-party discrimination affect hiring decisions in field settings, and to monitor how these beliefs evolve over time. Field experiments or panel data could test whether beliefs and firm behavior respond to information about others' discrimination. For example, quasi-experimental designs could use external shocks to beliefs of third-party discrimination (e.g., the Black Lives Matter protests or high-profile legal rulings, such as the U.S. 2023 ruling against affirmative action) to examine changes in third-party beliefs and contemporaneous changes in firms' hiring behavior. Data sources like those used in Hangartner et al. (2021), which include the universe of hiring decisions and information on a large recruitment platform in Switzerland, offer the opportunity to measure changes in discrimination following similar events.

Finally, as discussed above, this framework could apply beyond hiring and gender discrimination. There is extensive scope for future research to demonstrate the role of third-party beliefs on discrimination in other organizational and market contexts, to isolate where belief-biases might exist, and to test interventions to correct such biases, particularly where they exacerbate discrimination. While some evidence shows that misperceptions of third-party gender bias are widespread (Bursztyn et al., 2023), there is scope to investigate these misperceptions more extensively, for example regarding discrimination against other groups. The present work provides early evidence that such beliefs are both consequential and potentially inaccurate, but that they are subject to change, whether by external shock or informational intervention.

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Appendix

Table of Contents

A Natural Experiment Supplements

A.1 Sample Selection

I selected a set of 11 states where the election was predicted to be the more closely contested according to poll data. I started by selecting the 7 "swing states", each of which had a predicted margin of victory lower than 2 percentage points for each respective predicted winner. In order to create more variation in states and to increase the likelihood of variation in the state-level election winner in the case of a "landslide" victory in favor of one candidate within the swing states, I selected four additional states: two that predicted Donald Trump would win by a slightly higher margin and two that predicted Kamala Harris would win by a slightly higher margin. These were Ohio and Iowa in favor of Donald Trump and Virginia and Minnesota in favor of Kamala Harris. I specifically focused on states where poll data suggested the election would be closely contested in order to increase overall uncertainty about which candidate would win, allowing the election to be more informative about which candidate was most preferred in the state (Table C6 shows that there was a high degree of uncertainty about which candidate would win within the sample). Below is a table showing each state in the sample, the predicted margin of victory, and the eventual outcome for each.

Table A1: States in sample–poll predictions and election outcomes

	Polling Data		**Election Outcomes**	
State	Predicted Winner	Predicted Margin	Actual Winner	Actual Margin
Ohio	Trump	6.4	Trump	11.2
Iowa	Trump	4.3	Trump	13.2
North Carolina	Trump	1.5	Trump	3.2
Arizona	Trump	1.4	Trump	5.5
Georgia	Trump	1.4	Trump	2.2
Nevada	Trump	0.7	Trump	3.1
Pennsylvania	Trump	0.1	Trump	1.8
Wisconsin	Harris	1	Trump	1.1
Michigan	Harris	1.7	Trump	1.4
Minnesota	Harris	6.3	Harris	4.2
Virginia	Harris	8.8	Harris	5.7

Notes: Poll data was taken on the 2nd of November 2024 (2 days before the 2024 U.S. presidential election) from "270towin", a non-partisan platform that aggregates poll data across large polls predicting the U.S. 2024 election outcome.

Table A2: Final Sample Descriptive Statistics: Clients and Hiring Managers

	(1) Hiring Managers		(2) Client	
Variable	N	Mean/SD	N	Mean/SD
Age	412	42.00	85	41.56
		(12.24)		(15.14)
Male	412	0.52	85	0.56
		(0.50)		(0.50)
White	412	0.75	85	0.81
		(0.43)		(0.39)
Black or African-American	412	0.19	85	0.18
		(0.40)		(0.38)
Other Race	412	0.07	85	0.06
		(0.25)		(0.24)
College Graduate	412	0.56	85	0.52
		(0.50)		(0.50)
Has Advanced Degree	412	0.13	85	0.14
		(0.34)		(0.35)
Imputed HH Income ('000s)	412	74.90	85	69.32
		(47.38)		(47.81)
HH Income more than 75k	412	0.41	85	0.36
		(0.49)		(0.47)
Employed Full-time	328	0.57	63	0.44
		(0.50)		(0.50)
Unemployed	328	0.23	63	0.32
		(0.42)		(0.47)
Current student	337	0.11	72	0.08
		(0.32)		(0.28)
Trump won state	412	0.84	85	0.95
		(0.37)		(0.21)
Left-Right Political Scale	412	3.51	85	3.52
		(1.82)		(1.72)
Republican Party	412	0.30	85	0.27
		(0.46)		(0.45)
Democratic Party	412	0.50	85	0.48
		(0.50)		(0.50)
Other Party/Not Disclosed	412	0.21	85	0.25
		(0.41)		(0.43)
Has Real Hiring Experience	412	0.56	85	0.41
		(0.50)		(0.50)
Speaks multiple languages	412	0.10	85	0.08
		(0.30)		(0.28)
Speaks Spanish	412	0.05	85	0.04
		(0.22)		(0.19)
Completed Endline	490	0.84	100	0.85
		(0.37)		(0.36)

Notes: This table gives mean values (and standard deviations in brackets) for each variable for the final samples of hiring managers and clients used for analysis in the natural experiment, respectively (i.e., those that completed both baseline and endline surveys).

Table A3: Final and Baseline Sample Descriptive Statistics: Hiring Managers

	(1) Baseline		(2) Final Sample		T-test P-value
Variable	N	Mean/SD	N	Mean/SD	(1)-(2)
Age	497	41.28 (12.46)	412	42.00 (12.24)	0.38
Male	497	0.51 (0.50)	412	0.52 (0.50)	0.66
White	489	0.69 (0.46)	406	0.71 (0.46)	0.57
Black or African-American	489	0.19 (0.39)	406	0.17 (0.38)	0.48
Other Race	489	0.12 (0.33)	406	0.12 (0.33)	0.98
Employed Full-time	394	0.57 (0.50)	328	0.57 (0.50)	0.91
Unemployed	394	0.22 (0.41)	328	0.23 (0.42)	0.60
Current student	407	0.13 (0.33)	337	0.11 (0.32)	0.60
Trump won state	497	0.84 (0.37)	412	0.84 (0.37)	0.97
Republican Party	497	0.29 (0.45)	412	0.29 (0.45)	0.97
Democratic Party	497	0.51 (0.50)	412	0.52 (0.50)	0.91
Other Party/Not Disclosed	497	0.20 (0.40)	412	0.20 (0.40)	0.92
Speaks multiple languages	497	0.09 (0.29)	412	0.10 (0.30)	0.80
Speaks Spanish	497	0.05 (0.22)	412	0.05 (0.22)	0.96
F-test of joint significance (P-value)					0.99
F-test, number of observations					895

Notes: This table gives mean values (and standard deviations in brackets) for each variable for both hiring managers in the final sample (i.e., completing endline and baseline surveys) and hiring managers who only completed the baseline survey. The value displayed for t-tests and the F-test are p-values. The F-test result excludes employment and student status variables due to substantial missing observations (when we include all variables, and drop observations with missing values, the p-value for the F-test is 0.99) ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table A4: Final and Baseline Sample Descriptive Statistics: Clients

Variable	(1) Baseline N	(1) Baseline Mean/SD	(2) Final Sample N	(2) Final Sample Mean/SD	T-test P-value (1)-(2)
Age	100	41.16 (15.05)	85	41.60 (15.11)	0.84
Male	100	0.50 (0.50)	85	0.56 (0.50)	0.38
White	100	0.74 (0.44)	85	0.75 (0.43)	0.84
Black or African-American	100	0.13 (0.34)	85	0.13 (0.34)	0.99
Other Race	100	0.13 (0.34)	85	0.12 (0.32)	0.80
Employed Full-time	77	0.47 (0.50)	63	0.44 (0.50)	0.79
Unemployed	77	0.30 (0.46)	63	0.32 (0.47)	0.81
Current student	85	0.11 (0.31)	72	0.08 (0.28)	0.63
Trump won state	100	0.94 (0.24)	85	0.95 (0.21)	0.70
Republican Party	100	0.25 (0.44)	85	0.27 (0.45)	0.75
Democratic Party	100	0.55 (0.50)	85	0.52 (0.50)	0.66
Other Party/Not Disclosed	100	0.20 (0.40)	85	0.21 (0.41)	0.84
Speaks multiple languages	100	0.11 (0.31)	85	0.08 (0.28)	0.53
Speaks Spanish	100	0.06 (0.24)	85	0.04 (0.19)	0.44
F-test of joint significance (P-value)					0.99
F-test, number of observations					185

Notes: This table gives mean values (and standard deviations in brackets) for each variable for both clients in the final sample (i.e., completing endline and baseline surveys) and clients who only completed the baseline survey. The value displayed for t-tests and the F-test are p-values. The F-test result is using only variables without missing observations (when we include all variables, and drop observations with missing values, the p-value for the F-test is 0.99) ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

A.2 Baseline and Endline Survey Questions

Below I provide the belief and hiring task questions for clients and hiring managers in the natural experiment. Please find all (anonymized) materials for both experiments here on OSF.

A.2.1 Hiring Managers

Hiring Decisions [Baseline and Endline Surveys]:

- **[Incentivized]** Which worker do you choose? If you select the same worker the Client chose, you will receive $[1/2] if this question is selected for payment. You will receive $0 if you choose incorrectly. [Repeat 15 times]
  - Worker A
  - Worker B

Third-party Beliefs [Baseline and Endline Surveys]

- **[Incentivized]** Out of 15 worker pairs, how many times do you think clients from your state, [STATE], chose female workers on average? If you guess within 0.5 points of the correct average, you will earn $1 as a bonus payment if this part is selected for payment.
  - .[0–15 continuous sliding scale]

Below I will ask you some questions about your beliefs about Clients/people in your state of residence, [STATE]

- How do you think clients' (from your state) choices differed by worker gender?
  - They chose significantly more male workers
  - Slightly more male workers
  - No difference
  - Slightly more female workers
  - Significantly more female workers

- Imagine 100 randomly chosen people from your state were asked whether they agreed or disagreed with the following statement: "*Women should have the freedom to work outside of the home.*" How many people out of 100 do you think would agree with this?
  - Number that agreed [0–100 continuous sliding scale]
- How much would you say other people in your state generally agree with the following statement: "*When jobs are scarce, men should have more right to a job than women*"?
  - 1 – Strongly disagree
  - 2
  - 3
  - 4 – Neither agree nor disagree
  - 5
  - 6
  - 7 – Strongly agree

Productivity Beliefs [Endline Survey Only]:

- **[Incentivized]** Across all female workers, what do you think female workers' average score was for Quiz B? If you guess within 0.5 points of the correct average, you will earn $1 as a bonus payment if this part is selected for payment.
  - .[0–10 continuous sliding scale]
- **[Incentivized]** Across all male workers, what do you think male workers' average score was for Quiz B? If you guess within 0.5 points of the correct average, you will earn $1 as a bonus payment if this part is selected for payment.
  - .[0–10 continuous sliding scale]

State Election Beliefs [Baseline Survey Only]

- **[Incentivized]** Which candidate will win the 2024 U.S. Presidential Election in your state, [STATE]? If you guess correctly, you will earn $1 as a bonus payment if this part is selected for payment.

  - Kamala Harris
  - Donald Trump
  - Other candidate

- **[Incentivized]** By what percentage margin do you think this candidate will win in your state (ie. % more of the total votes in the state than the next candidate)? If you guess within 3 percentage points of the true margin, you will earn $1 as a bonus payment if this part is selected for payment.

  - Percentage victory margin [0–50 continuous sliding scale]

- How sure are you that this candidate will win in your state?

  - 1 – Very unsure
  - 2
  - 3
  - 4 – Not particularly sure
  - 5
  - 6
  - 7 – Completely sure

National Election Beliefs [Baseline Survey Only]

- **[Incentivized]** Which candidate do you think will win the 2024 US Presidential Election Nationally (ie. will win the Electoral College)? If you guess correctly, you will earn $1 as a bonus payment if this part is selected for payment.

  - Kamala Harris
  - Donald Trump
  - Other candidate

- **[Incentivized]** By what margin of total Electoral College votes/seats do you think this candidate will win (ie. how many more votes/seats than the next candidate)? If you guess within 10 electoral seats of the true margin, you will earn $1 as a bonus payment if this part is selected for payment.
  - Electoral votes victory margin [0–150 discrete (integer) sliding scale]
- How sure are you that this candidate will win the Electoral College?
  - 1 – Very unsure
  - 2
  - 3
  - 4 – Not particularly sure
  - 5
  - 6
  - 7 – Completely sure

A.2.2 Clients

Selection Decisions [Baseline and Endline Surveys]:

- **[Incentivized]** Which worker do you choose? If you select the worker with the highest Quiz B score, you will receive $[1/2] if this question is selected for payment. You will receive $0 if you choose incorrectly. [Repeat 15 times]
  - Worker A
  - Worker B

Productivity Beliefs [Baseline and Endline Surveys]:

- **[Incentivized]** Across all female workers, what do you think female workers' average score was for Quiz B? If you guess within 0.5 points of the correct average, you will earn $1 as a bonus payment if this part is selected for payment.

  - .[0–10 continuous sliding scale]

- **[Incentivized]** Across all male workers, what do you think male workers' average score was for Quiz B? If you guess within 0.5 points of the correct average, you will earn $1 as a bonus payment if this part is selected for payment.
  - .[0–10 continuous sliding scale]

A.3 The Worker Pool

100 Prolific participants completed two Tasks, which were tests of their basic math, finance, and history knowledge. Each task, Quiz A and Quiz B included 10 randomly-selected questions from these topics. After completing the tasks, workers answered several demographic questions, which included their self-reported gender and highest level of education. Please find all task materials here on OSF.

To construct worker pairs for this study, workers were randomly selected from the full pool of 100 workers and paired together. I then reduced the number of pairs, over-sampling: (1) mixed-gender pairs, (2) pairs where workers were more similarly qualified according to the signals provided to decision makers (i.e., where their Quiz A scores and highest level of education differed by less than 2 units), and (3) "ambiguous" pairs, where one candidate dominated the other on one signal (i.e., on their highest level of education or on their score on Quiz A) but not on the other (this was in order to provide decision makers with "cover" to choose the candidate they truly believe is better; see Barron et al. (2024) for details of this procedure). These strategies were predominantly used to increase statistical power to more closely mimic choices between similarly qualified job candidates, and to reduce plausible social desirability bias. The final set of worker pairs had similar characteristics to the full pool of workers (they did not significantly differ with the full worker pool on any characteristic), and male and female workers in the final pairs did not significantly differ from each other on any characteristic (see Table A5 for a summary of worker pairs). For each hiring manager and client, 15 pairs were randomly selected from the pool of worker pairs, and randomly ordered.

There were two additional exceptions to full randomization of pair selection and order, both of which were designed to reduce social desirability bias. First, the first four pairs decision makers were given were always the same, though they were randomly ordered within this set of four. These four pairs were designed to be relatively "easy decisions", where one candidate had both a higher Quiz A score and higher education (i.e., strictly dominated the other candidate with signals other than gender). Moreover, I also over-sampled same-sex pairs within this set of four (both male and female). This was in order for decision makers to select at least some male *and* female workers in the early stages of the survey. This would thus provide decision makers with "cover" for choosing workers they truly believed would maximize their payoffs without appearing discriminatory. The second exception is that the 15th and final pair always included a randomly selected male-female pair where each worker had the same education level and Quiz A score. Given that gender is the only way to distinguish between these workers, it makes gender more salient, and reduces the potential cover that decision makers had. I thus placed this last so that this salience and lack of cover did not allow them to make compensating decisions afterwards.

Table A5: Stimuli Summary: Worker Characteristics by Worker Gender in Final Pairs

	Female	**Male**
Quiz Scores (Mean)		
Quiz A Score	3.44	3.59
Quiz B Score	3.52	3.12
Highest Education (Percentage)		
High School	0.148	0.148
Some College	0.148	0.074
College Graduate	0.444	0.593
Advanced Degree	0.259	0.185
Imputed average years of education	15.89	15.81
Pair features		
Dominant	0.500	0.500
Worker A	0.519	0.481
N	27	27

Notes: This table provides mean values for characteristics of male and female workers included in the final set of worker pairs in the hiring task (i.e., the final set of workers displayed to both clients and hiring managers during the main phase).

A.4 Analysis Strategy

For my primary analysis, I exploit both cross-state variation in which candidate won the 2024 National Election, and variation in the "surprisingness" of the election results determined by participants' baseline beliefs about which candidate would win the election and the winning margin. I estimate the below set of specifications using ordinary least squares (OLS) regression, following my primary preregistered analyses.

$$Y_i = \beta_0 + \beta_1 TrumpState_i + \beta X_i + \epsilon_i \tag{1}$$

$$Y_i = \beta_0 + \beta_1 ElectionSurprise_i + \beta X_i + \epsilon_i \tag{2}$$

$$Y_i = \beta_0 + \beta_1 ElectionSurpriseDelta_i + \beta X_i + \epsilon_i \tag{3}$$

Y_i represents the outcome of interest for hiring manager i. In my primary preregistered analysis, this is the *change* in the weight placed on gender by participant i when hiring workers. This measures the change in the likelihood of hiring a female worker relative to a male worker, holding other qualifications (education and aptitude signal) constant. To acquire these gender weights, I regressed each individual participants' choices for the 15 worker pairs they observed on each worker pair's gender difference, difference in Quiz A scores, and difference in their highest level of education. This allowed me to determine the effect of gender on each decision maker's choices, controlling for other differences in worker qualifications.[20] For all participants in the final sample, I observe hiring choices both immediately before and soon after the election. I derived these "gender weights" separately for each round, and found the difference by subtracting the weight at baseline from the weight at endline. Positive values for Y_i represent an increase in the likelihood of hiring female workers after the election compared to before, holding worker qualifications constant, and negative values represent a decrease in the likelihood of hiring female workers. The first equation measures the effect of being a resident in a state where Trump won the highest vote share (the

[20]This provided a more precise measure of gender discrimination than the proportion of female workers a participant hired, but is highly correlated with this measure ($r = 0.89$). A weight of 0 implies that the participant does not use gender to make decisions, positive values imply favoring female workers, and negative values imply favoring male workers.

variable $TrumpState$) on Y_i. Given the research described in the manuscript in Section 2.1.5, Trump winning in a given state could have affected beliefs of others' discrimination in that state and hiring managers' own discrimination. By using a difference variable as the primary outcome, I remove the effect of time invariant features of participants in each state, and focus on the effect of a Trump victory on changes in female worker hiring. This represents the primary "manipulation" in the study. While it is unlikely that substantive changes in beliefs or preferences relevant to gender discrimination (and unrelated to the election outcome) occurred in the 12-17 days between baseline and endline measurement (given its brevity and the salience of the election), I run this specification with a series of different control variables X[21] and conduct checks described in Section 2.2.2 for robustness.

In the second equation, I test the effect of a "surprising" election outcome on changes in female worker hiring. The variable $ElectionSurprise_i$ uses within-state variation in "surprisingness" of the election across hiring managers by exploiting their baseline beliefs about which candidate would win the highest vote share in their state. Trump winning in a manager's state plausibly causes them to update their beliefs more if the manager didn't expect this victory. The same should be true, but with a directionally opposite effect, if the manager did not expect Kamala Harris to win and she did. $ElectionSurprise_i$ is coded as 1 if Trump won in participant i's state and the participant predicted Harris would win, 0 if the participant correctly predicted which candidate would win, and -1 if Harris won and they predicted Trump would win. The final preregistered specification, Equation 3, makes use of variation in the continuous "margin" of surprisingness of the election outcome. This measure, reflected by the variable $ElectionSurpriseDelta_i$, captures the discrepancy between Trump's winning (or losing) margin and the hiring manager's prediction for Trump's winning (losing) margin, with positive values indicating that Trump received a larger relative share of the vote than the manager expected and negative values meaning that Trump received a smaller relative share of the vote than the manager predicted. More positive values thus indicate a more surprising outcome in favor of Trump. For instance, if a hiring manager lived in Michigan, a state where Trump won the vote share by a margin of 1.4%, and predicted a

[21] Control variables include dummy variables for black and white race, male gender, continuous variables for age, age squared, and incentivized beliefs about the relative male and female worker productivity, and fixed effects for political party affiliation, income, and highest education level.

Harris victory of 2%, this value would take 3.4, reflecting the discrepancy between their prediction and the outcome in the direction of a Trump victory. If the participant had instead predicted a Trump victory of 2%, this would take a value of -0.6.[22] For robustness, I compare this result to an alternative that uses the manager's self-reported level of "certainty" in their estimate of which candidate would win (on a 7-point Likert scale) to also measure the effect of the degree of surprise of the election result in favor of Trump (these results are given in Figure A1).

I conduct the same set of analyses to measure the effect of the election on managers' beliefs about client discrimination (for clients in their state), running the same OLS regressions using the change in managers' incentivized beliefs about the proportion of female workers clients selected as Y_i. Conditional on this "first-stage" result, I then conduct a two-stage-least-squares (2SLS) instrumental variable regression, using the above independent variables ($TrumpState$, $ElectionSurprise$, and $ElectionSurpriseDelta$) as instruments for the change in beliefs about client discrimination, to test for the causal effect of these beliefs on hiring choices.

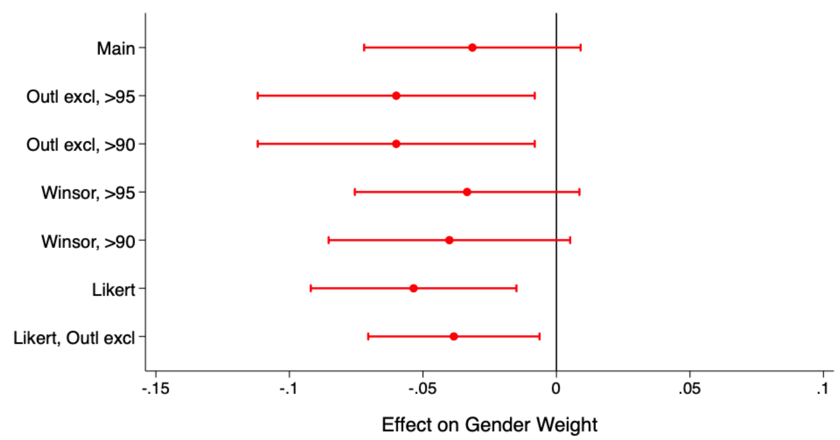
[22]This is the key interpretation for this variable. However, in regression output, purely for presentation purposes so that coefficients are visible, I scale this variable dividing by 50 (i.e., the coefficients represent a 50% deviation between predicted and actual margin of victory)

A.5 Additional Analyses

A.5.1 Robustness of Marginal "Surprise" Effect in Election

The effect of election "surprise" on female worker hiring was only significant for a "categorical" surprise (whether or not the participant incorrectly predicted which candidate won) but not for a "marginal" surprise (the discrepancy between the expected margin of victory and the actual margin of victory), even though belief updating should theoretically be sensitive to both. To check for the robustness of the effect of the continuous measure of the elections surprisingness, I account for outliers in participants' expectations about the margin of victory. Given the closeness of the election in the states in which I investigate, large expected margins of victory might also reflect participants being uninformed or responding inattentively (even though these measures were incentivized, and participants passed a set of attention checks). Indeed, the largest actual margin of victory across these states was 13 percentage points (pp). However, 20% of participants indicated an expected margin of victory higher than 20 pp, and 6% said that it would be higher than 40 pp. This suggests that including these participants might add noise to the analysis. I account for outliers by winsorizing and dropping outliers at different thresholds, and test an alternative measure for how "surprising" a result was (using a 7-point Likert measure of their level of confidence in their prediction of the state election winner). I find that excluding outliers for the belief of margin of victory shifts this coefficient to significance, while the alternative measure of marginal surprisingness also shows a significant negative effect (Figure A1). In spite of this significant result, however, the pattern seems to largely be driven by the "binary surprise" of which candidate won, rather than by the margin of victory. First, the coefficient on the specification with the largest effect shows that a surprise in the election outcome of 20 percentage points in favor of Trump (a very large belief shock) would lead to only a 2.5 percentage point reduction in female worker hiring, compared to a 3.5 pp reduction from failing to predict the winning candidate. Moreover, when I include the binary surprise variable in a regression with the marginal surprise variable, the effect of the marginal surprise disappears.

Figure A1: The Effect of Marginal Election Surprise on Hiring: Robustness



Notes: Red dots represent β_1 coefficients on the $ElectionSurpriseDelta_i$ variable from Equation 3 in Supplement A.4, representing the effect of the deviation between managers' baseline beliefs about the election outcome and the actual election outcome. Red lines represent 95% confidence intervals of the coefficient. Each dot and line represent the coefficient for a different operationalization of $ElectionSurpriseDelta_i$ using different strategies to account for outliers, which were (from top to bottom): (i) the main specification (defining $ElectionSurpriseDelta_i$ as the deviation between thee eventual margin of victory for the winning candidate and managers' predicted margin of victory); (ii) same definition but excluding observations where the absolute value of $ElectionSurpriseDelta_i$ was greater than the 90th percentile; (iii) same definition but excluding observations where the absolute value of $ElectionSurpriseDelta_i$ was greater than the 95th percentile; (iv) same definition but winsorizing at the 95th percentile; (v) same definition but winsorizing at the 90th percentile; (vi) using an alternative measure of $ElectionSurpriseDelta_i$, which is the interaction of the variable $ElectionSurprise_i$ (defined in Supplement A.4; the direction of the surprise in terms of a Trump vs. Harris victory) and the managers' level of certainty (measured on a 7-point Likert scale) that their prediction of which candidate would win was correct; conceptually, this implies that the value is greater when they were incorrect *and* more certain, and thus the election should have been more surprising; (vii) the same definition excluding observations with levels of certainty of 1 or 7 (completely uncertain or completely certain).

A.5.2 Non-incentivized Belief Changes

In addition to the primary incentivized belief measure about third-party discrimination (i.e., hiring managers' estimates about the proportion of women that clients' selected), I included two measures taken from the World Values Survey and the Gallup World Poll, respectively, to elicit hiring managers' beliefs about others' more general attitudes towards female labor force participation. For beliefs about others' more general attitudes, the results are more mixed than for the main incentivized belief measure. The Gallup World Poll measure, which asks managers to predict the number of people out of 100 randomly selected people from their state who agree with the statement "Women should have the freedom to work outside of the home", did not change their beliefs as predicted (Table C8, and Table C9). However, the World Values Survey measure, which is a Likert measure of how much managers think other people in their state agree with the statement "When jobs are scarce, men should have more right to a job than women", followed the same pattern as the incentivized measure of beliefs about client discrimination above. That is, both Trump winning a state and a surprising Trump victory in that state significantly predicted increases in expectations that others agreed jobs should be reserved for men.

A.6 Social Desirability Bias

While I cannot rule out the presence of socially desirable responding, several features of the study design and observed data limit concerns that demand effects or social desirability bias affected the main findings. First, for both experiments, all responses were anonymous and decisions and belief estimates were incentivized, with accuracy bonuses that were large relative to participants' base pay (a $1 or $2 bonus for a correct randomly selected hiring choice, and 1$ for a correct randomly selected belief estimate, compared to $1-1.30 base pay). Reflecting this, participants were highly responsive to workers' productivity signals in their choices. Worker signals, (i.e., their highest level of education and score on Quiz A) accounted for 31.5-34.9% of the variation in hiring managers' choices across both experiments. Furthermore, when one worker was strictly dominant in a worker pair (i.e., had a higher level of education and a higher score on Quiz A), hiring managers chose that worker 95.3% of the time, irrespective of worker gender.

Second, the hiring task was specifically constructed to reduce the salience of gender and to provide decision makers with "cover" for selecting either male or female workers in each decision. Given that the survey was already anonymous, this should have further reduced participants' reputational concerns about responding in certain ways. For instance, in the natural experiment, gender was not mentioned at any point outside of worker profiles and belief elicitation questions.[23] Moreover, the hiring task used in both experiments included several features to limit the salience of selecting workers on the basis of gender: (i) I over-sampled worker pairs where one worker was dominant on one signal but not on the other, allowing more opportunity to justify worker selection on the basis of productivity signals (Barron et al., 2024; De Quidt, Haushofer, & Roth, 2018);[24] (ii) the first several worker pairs included a higher proportion of both same-sex and strictly dominant pairs, encouraging decision makers to select male *and* female workers and to focus on productivity signals rather than gender in the early stages of the task; (iii) male-female worker pairs where both workers were identical except for gender, and thus where gender was more salient, were delayed until the

[23] This was the same in the controlled experiment except for treatment conditions where the Quiz B score distribution broken down by worker gender was provided: importantly, none of the primary hypothesis tests in this paper rely on comparisons between participants that did vs. did not receive this information.

[24] As discussed in Barron et al. (2024), this allows decision makers to justify the choice of one worker over the other based on the weight they place on signals of productivity, rather than on gender. For instance, if a female worker in a pair is more educated than her male counterpart but has a lower score on Quiz A, choosing the male worker can be justified by placing more weight on Quiz A scores, claiming this to be more diagnostic of productivity.

15th and final pair to limit spillovers from this choice to other choices. Together, these features should have increased decision makers "cover" for choosing workers they believed would maximize their payoffs and limiting their concerns about appearing discriminatory.

Finally, while social desirability might pose a concern for the levels of female worker hiring in the studies, it should not necessarily drive the differences in hiring observed between treatment conditions. In the controlled experiment, the main preregistered hypothesis tests involved differences in managers' choices and beliefs between the *Clients Debiased* and the *Clients as-is* conditions. Both treatment groups received the same information about workers' Quiz B scores by worker gender, only differing in whether they were told clients saw "exactly the same" vs. "did not see" this information. As such, gender was similarly salient for each group and socially desirable responding should not have driven differences in responses across conditions.[25].

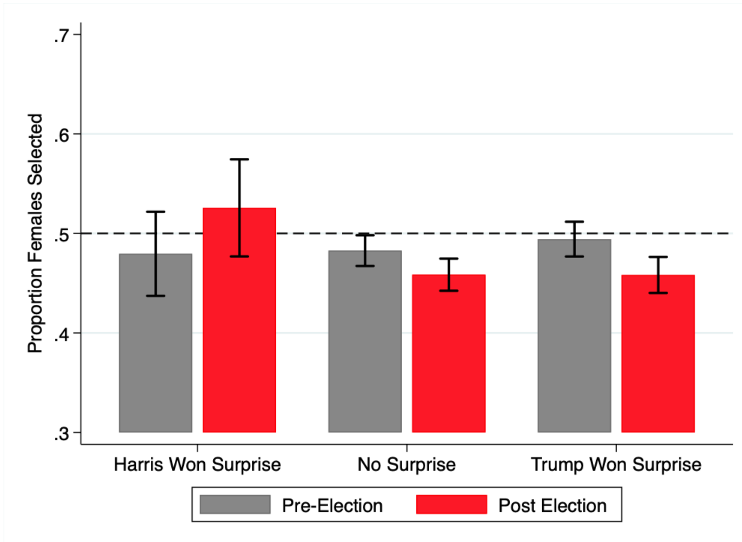
In the natural experiment, information about the experimenter and the task instructions and format were the same both before and after the election and for all managers. As such, it is unlikely that experimenter demand would have driven *changes* in manager choices over the election, particularly differentially across states and across managers with different baseline beliefs about the election.

The comparison between hiring managers and clients was potentially more vulnerable to social desirability bias. Hiring managers' decisions and beliefs were based on potential discrimination by *others*, possibly reducing their concerns about appearing discriminatory, as they could justify their actions by appealing to others' (rather than their own) discriminatory beliefs. Clients thus might have faced relatively more pressure to respond in socially desirable ways, as their choices reflected their own beliefs. While I cannot rule out this possibility, several features limit this concern; (i) The previously described features should limit social desirability concerns for *all* participants; (ii) hiring managers were aware of the population from which clients were selected and clients' decision-problem, and were incentivized to accurately predict client choices: they could thus have incorporated client social desirability bias in their estimates; (iii) finally, if clients were more susceptible to social desirability than hiring managers, they should have been more biased in their choices; however, clients chose the higher performing worker more often than hiring managers did.

[25]Gender was potentially more salient for participants in these two conditions than those assigned to the *Control* condition (who were not provided the Quiz B score distributions by worker gender). However, the primary hypothesis tests do not focus on comparisons between either condition and the *Control* condition.

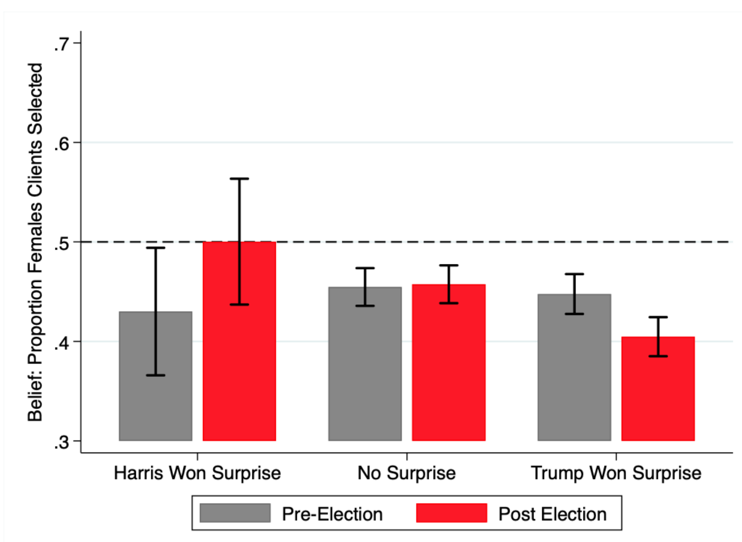
B Natural Experiment - Additional Figures

Figure B2: Manager Hiring: Proportion of Female Workers Selected by Categorical Surprise Baseline and Endline



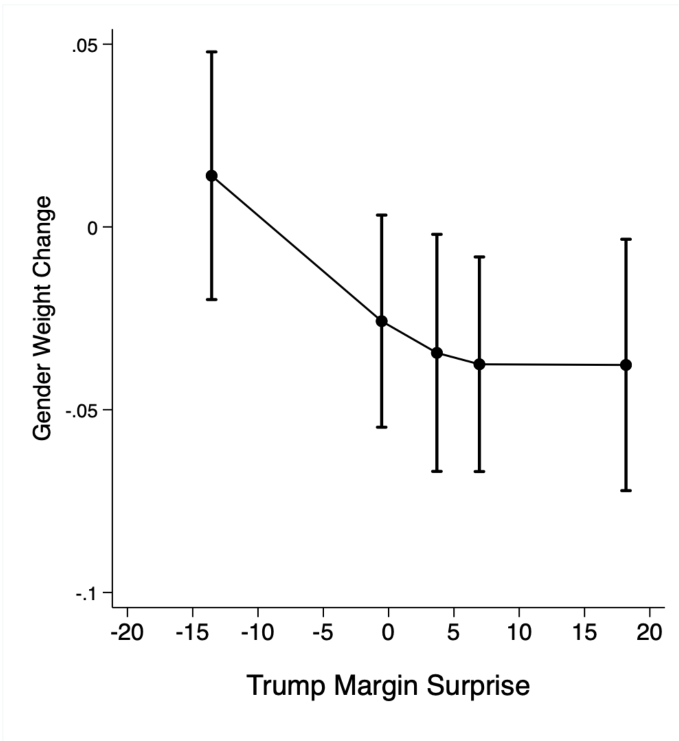
Notes: Bars show the mean proportion of female workers hired by hiring managers, separately for hiring managers that i) failed to predict a Harris victory in their state (i.e., Harris won, but they predicted Trump would win), ii) correctly predicted the winning candidate, iii) failed to predict a Trump victory in their state. Grey bars show these means for pre-election (baseline) measures and red bars show means for post-election (endline) measures. The dotted line represents 50% male and female workers hired. Error bars represent 95% confidence intervals of the means.

Figure B3: Manager Beliefs about the Proportion of Female Workers Clients Selected by Categorical Surprise; Baseline and Endline



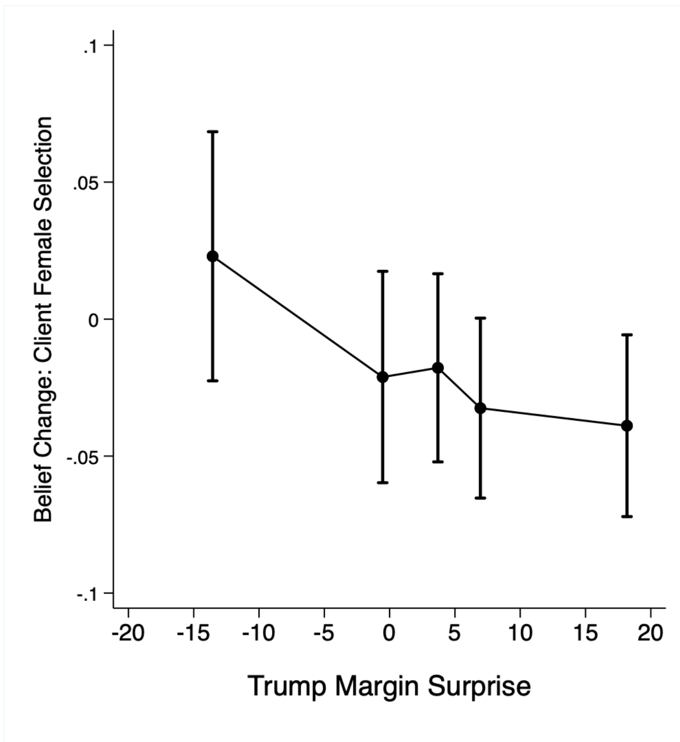
Notes: Bars show the mean hiring manager belief about the proportion of female workers clients selected, separately for hiring managers that i) failed to predict a Harris victory in their state (i.e., Harris won, but they predicted Trump would win), ii) correctly predicted the winning candidate, iii) failed to predict a Trump victory in their state. Grey bars show these means for pre-election (baseline) measures and red bars show means for post-election (endline) measures. The dotted line represent beliefs of 50% male and female workers selected by clients. Error bars represent 95% confidence intervals of the means.

Figure B4: Manager Hiring: Change in Gender Weights by Quintile of Election Surprisingness (Measured by Predicted Margin of Victory)



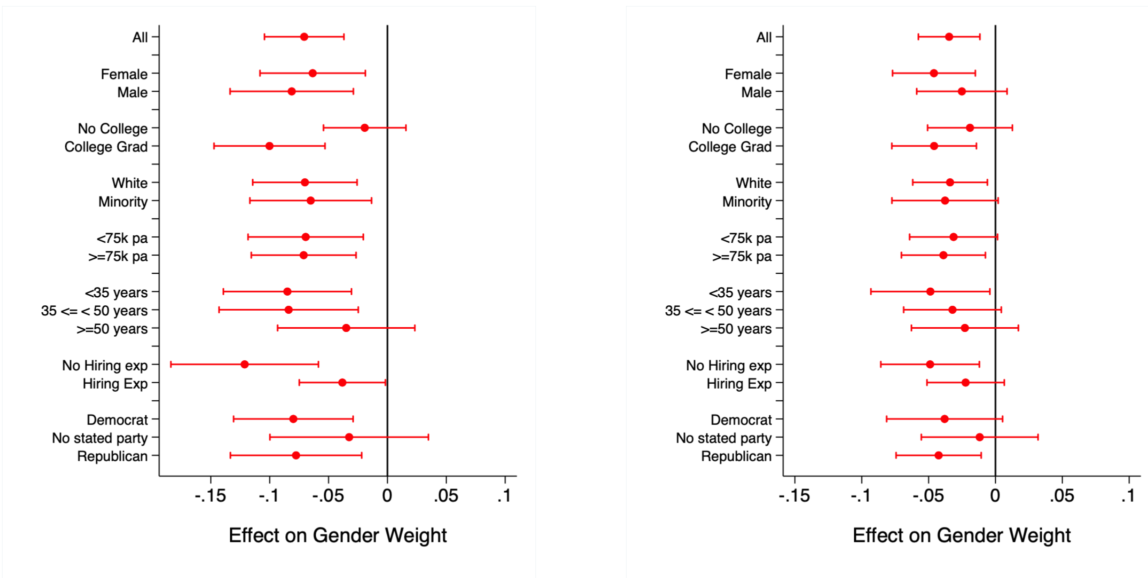
Notes: Dots represent the mean change in the decision weight that hiring manager's placed on gender over the election (positive values indicate weighing female gender more positively after the election than before, and negative weights indicate weighing female gender more negatively after the election). Each dot is plotted for a separate quintile of the surprisingness of the election outcome, measured by the difference between the actual and manager's predicted margin of victory in favor of Donald Trump in the election. Positive values indicate cases where Trump received more votes than managers predicted, and negative values where Trump received fewer votes than managers predicted. Error bars represent 95% confidence intervals of the means.

Figure B5: Change in Manager Beliefs about the Proportion of Female Workers Clients Selected by Quintile of Election Surprisingness (Measured by Predicted Margin of Victory)



Notes: Dots show the mean the change in hiring manager's incentivized beliefs about client discrimination, calculated by subtracting the baseline from the endline belief measure (of beliefs of the proportion of workers clients selected that are female). Each dot is plotted for a separate quintile of the surprisingness of the election outcome, measured by the difference between the actual and manager's predicted margin of victory in favor of Donald Trump in the election. Positive values indicate cases where Trump received more votes than managers predicted, and negative values where Trump received fewer votes than managers predicted. Error bars represent 95% confidence intervals of the means.

Figure B6: Change in Gender Weight in Hiring by Belief Shock: Heterogeneity

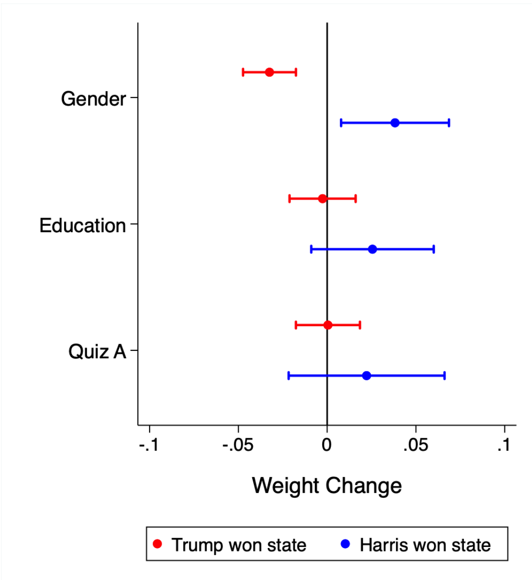


(a) by "Trump Won"

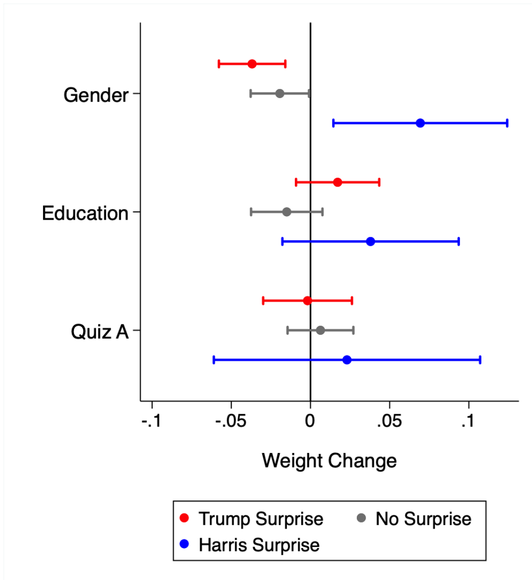
(b) by "Election Surprise"

Notes: This figure shows heterogeneity in the effect of the election on changes in the decision weight placed on female workers for different "belief shocks". In Panel (a) each dot represents the β_1 coefficient on $TrumpState_i$ from Equation 1 in the Supplement A.4 estimated separately for different subgroups of hiring managers. This tests the effect of the election outcome (i.e., being in a state where Trump vis-a-vis Harris won the election) on the change in the likelihood of hiring a female worker (represented by the weight placed on the worker being female in decisions) for different subsets of managers. The y-axis describes the specific subgroup of managers for which Equation 1 was estimated, estimating this separately based on manager gender, highest level of education, race, income, age, previous hiring experience, and political party affiliation respectively. In Panel (b) each dot represents the β_1 coefficient on $ElectionSurprise_i$ from Equation 2 in the Supplement A.4 estimated separately for the same subgroups of hiring managers. In both panels, lines represent 95% confidence intervals of the estimated coefficients.

Figure B7: Change in Decision Weights by Belief Shock



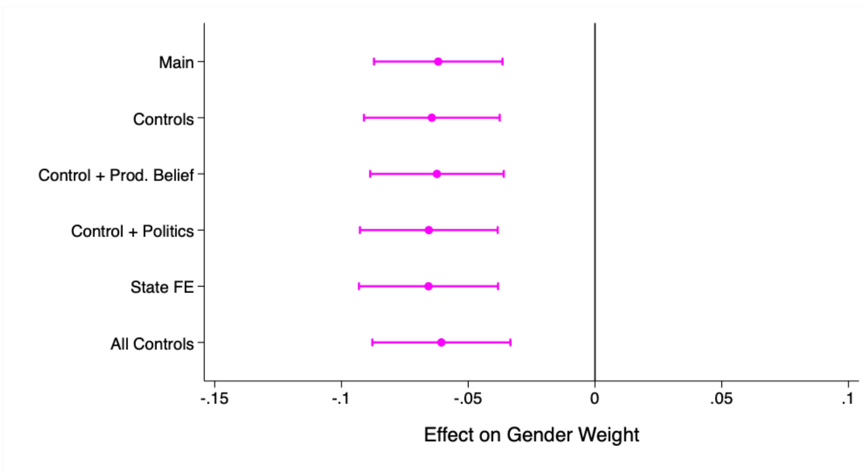
(a) by "Trump Won"



(b) by "Election Surprise"

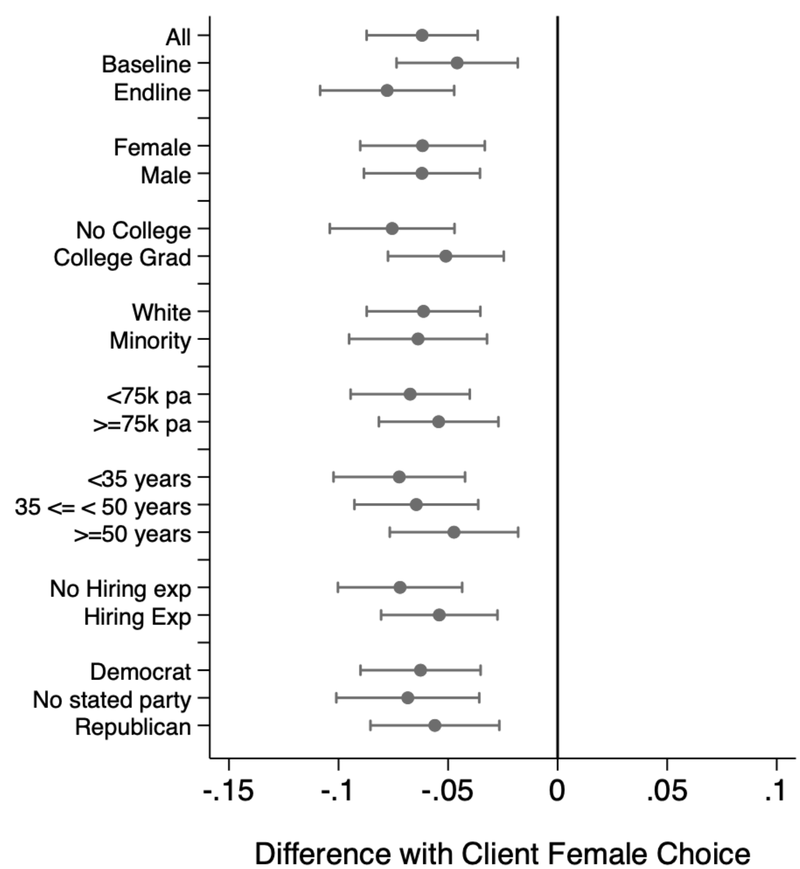
Notes: In both panels, each dot represents the mean change over the election period (i.e., between baseline and endline surveys) in the decision weights hiring manager placed on each worker attribute when selecting workers in the hiring task. In both panels, lines represent 95% confidence intervals of the mean change in decision weight. In Panel (a), this is split between managers in states where Trump won the election (the red dots) and managers in states where Harris won the election (the blue dots). In Panel (b) this is split between managers who failed to predict Trump winning in their state ("Trump Surprise"; red dots), managers who correctly predicted the winning candidate in their state ("No Surprise"; grey dots), and who failed to predict Harris winning in their state ("Harris Surprise"; blue dots).

Figure B8: Robustness: Difference between Hiring Manager and Client Gender Weight



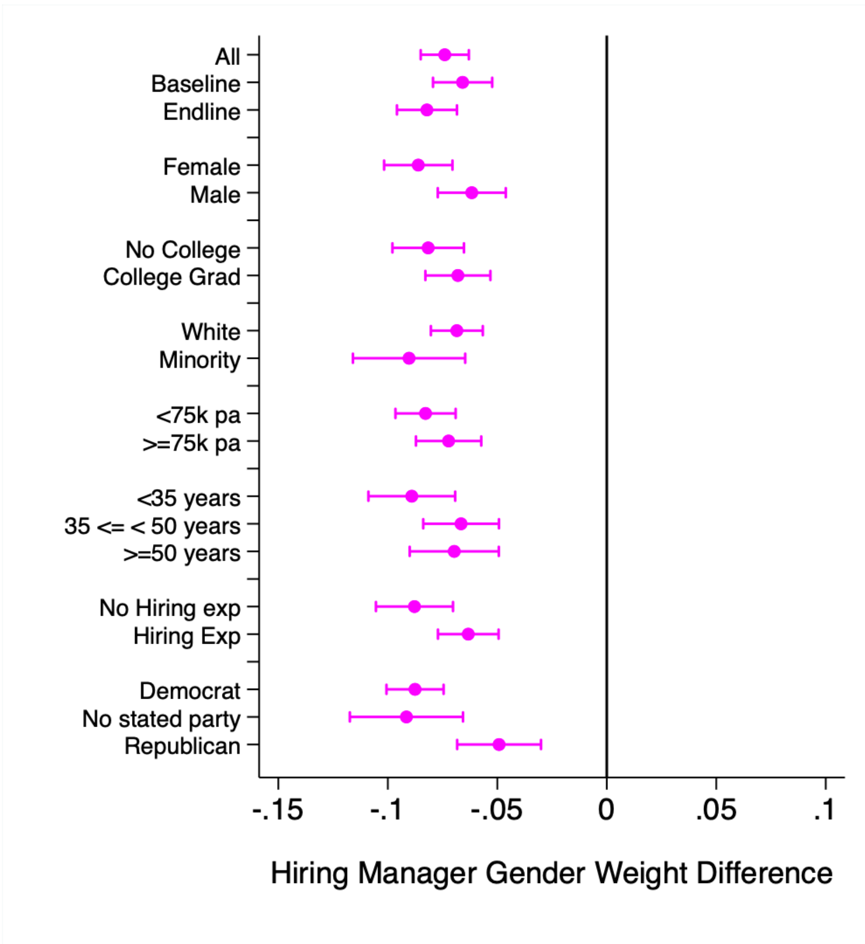
Notes: Each dot represents regression estimated difference between hiring managers and clients in the likelihood of hiring a female worker (holding other worker signals constant). The coefficient is specifically on the variable $HiringManager_i$ from an OLS regression of the decision weight placed on the worker being female (the dependent variable) on an indicator that the decision maker was a hiring manager ($HiringManager_i = 1$) rather than a client ($HiringManager_i = 0$), and a vector of control variables X_i. This represents the primary pre-specified analysis to test the hypothesis that hiring managers and clients discriminate against female workers differently. Each row represents the results of a specification with a different set of control variables. Control variables vary by specification as indicated, but the base controls (rows 2-6) include dummy variables for black and white race, male gender, college educated (or not), and continuous age and age squared. "Prod. belief" indicates incentivized beliefs about relative male and female worker productivity, measured by predicted task scores; "Politics" refers to political party affiliation fixed effects, and "All Controls" represent all the above control variables, and additionally income and highest education level fixed effects. Lines represent 95% confidence intervals of the estimated coefficients. As preregistered, all regressions include pooled observations (i.e., estimated decision weights for each participant) for both clients and hiring managers and at both baseline and endline.

Figure B9: Heterogeneity: Difference between Hiring Manager Beliefs about Client Choices and Clients' Actual Choices



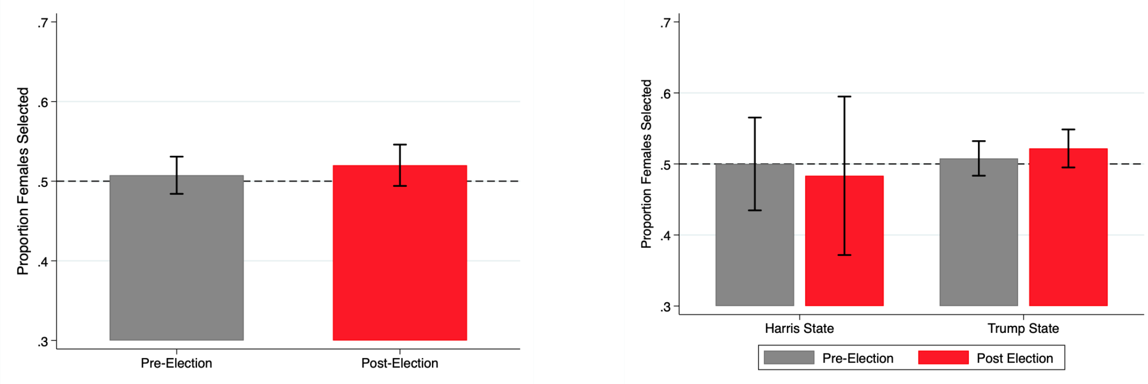
Notes: Grey dots represent mean differences between hiring managers' beliefs about the proportion of female workers that clients selected, and the proportion of female workers that clients actually selected (the "belief discrepancy"). Lines represent the 95% confidence intervals for these difference estimates. Each row represents these estimates for separately different subsets of participants based on participant gender, highest level of education, race, income, age, previous hiring experience, and political party affiliation respectively.

Figure B10: Heterogeneity: Difference between Hiring Manager and Client Gender Weight



Notes: Each dot represents regression estimated difference between hiring managers and clients in the likelihood of hiring a female worker (holding other worker signals constant). The coefficient is specifically on the variable $HiringManager_i$ from an OLS regression of the decision weight placed on the worker being female (the dependent variable) on an indicator that the decision maker was a hiring manager ($HiringManager_i = 1$) rather than a client ($HiringManager_i = 0$). Lines represent the 95% confidence intervals for these coefficient estimates. Each row represents these coefficients estimated separately (i.e., on a subset of participants) based on participant gender, highest level of education, race, income, age, previous hiring experience, and political party affiliation respectively.

Figure B11: Clients' Selection: Proportion of Female Workers Selected

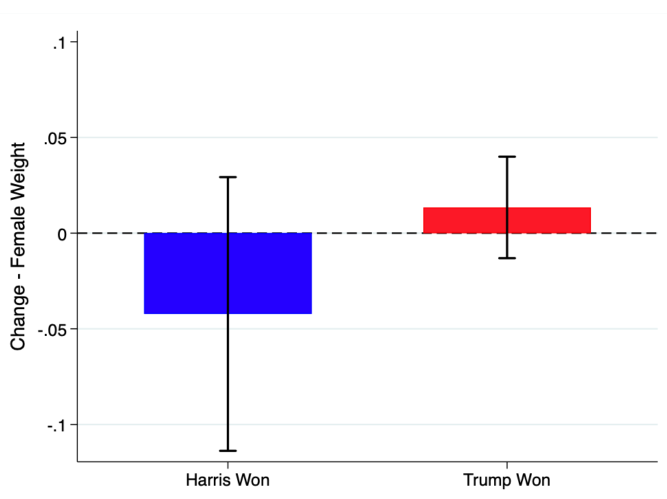


(a) Pre and Post Election

(b) Pre-post by State Election Result

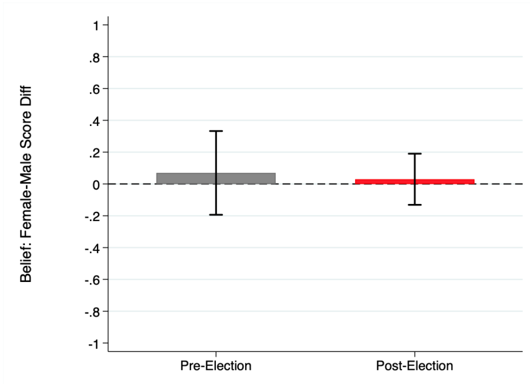
Notes: Panel (a) shows the mean proportion of female workers selected by clients before and after the election, across all clients. Panel (b) shows the mean proportion of female workers selected by clients, separately for i) U.S. states in which Trump won the highest share of votes in the 2024 National Election and states in which Harris won the highest share, and ii) pre-election (baseline) and post-election (endline) measures. The dotted line represents 50% male and female workers selected. Error bars represent 95% confidence intervals of the means.

Figure B12: Clients' Change in Decision Weight on Gender

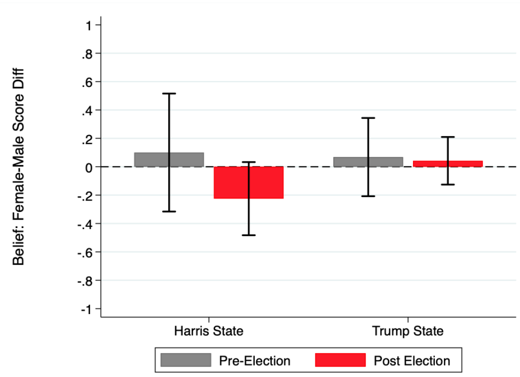


Notes: Bars show the mean change in the decision weight that client's placed on gender over the election (positive values indicate weighing female gender more positively after the election than before, and negative weights indicate weighing female gender more negatively after the election). These weight changes represent the change in the likelihood of selecting female workers vis-a-vis male workers, holding worker qualifications constant. Error bars represent 95% confidence intervals of the means.

Figure B13: Clients' Predicted Worker Scores: Female minus Male Scores



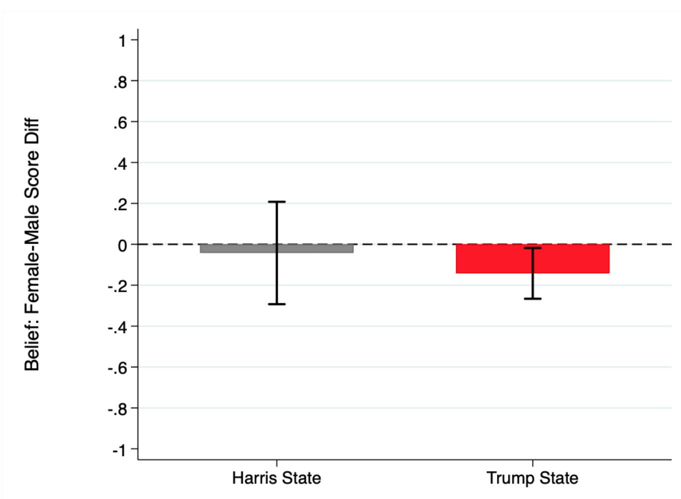
(a) Pre and Post Election



(b) Pre-post by State Election Result

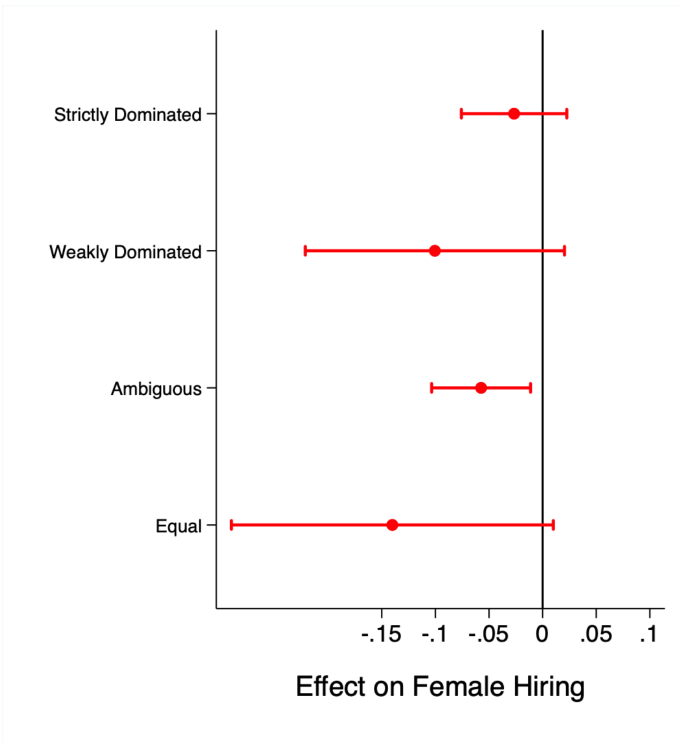
Notes: Panel (a) shows the mean difference between clients' predicted Quiz B score for female workers and their predicted Quiz B score for male workers before and after the election, across all clients. Panel (b) shows the same mean difference for clients, separately for i) U.S. states in which Trump won the highest share of votes in the 2024 National Election and states in which Harris won the highest share, and ii) pre-election (baseline) and post-election (endline) measures. Error bars represent 95% confidence intervals of the means.

Figure B14: Hiring Managers' Predicted Worker Scores at Endline: Female minus Male Scores, by Election Winner in State



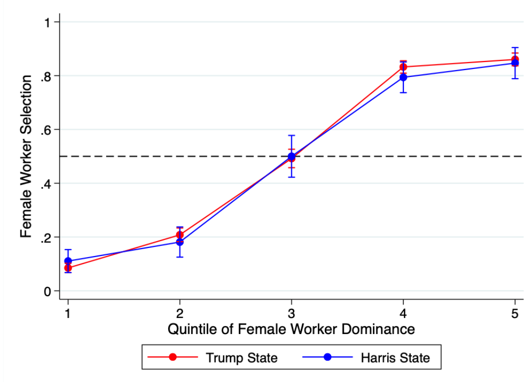
Notes: Bars show the mean difference between hiring managers' predicted Quiz B score for female workers and their predicted Quiz B score for male workers in the endline survey (post-election), separately for U.S. states in which Trump won the highest share of votes in the 2024 national election and states in which Harris won the highest share. Error bars represent 95% confidence intervals of the means.

Figure B15: The Effect of "Trump State" on Hiring, by Type of Worker Pair

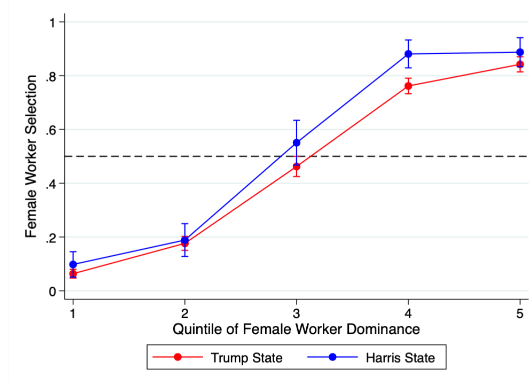


Notes: Each dot represents the β_1 coefficient on $TrumpState_i$ from Equation 1 in the Supplement A.4 estimated separately for different Y_i. Y_i represents the decision weight placed on the worker being female, holding fixed worker highest level of education and score on Quiz A (i.e., their signals) for different types of worker pairs. This tests the effect of the election outcome (i.e., being in a state where Trump vis-a-vis Harris won the election) on the change in the likelihood of hiring a female worker in worker pairs where: (i) one worker strictly dominates (based on highest education and Quiz A score), (ii) one worker weakly dominates, (iii) one worker is higher on one signal and lower on the other, and (iv) where worker signals are equal. Lines represent 95% confidence intervals of the estimated coefficients.

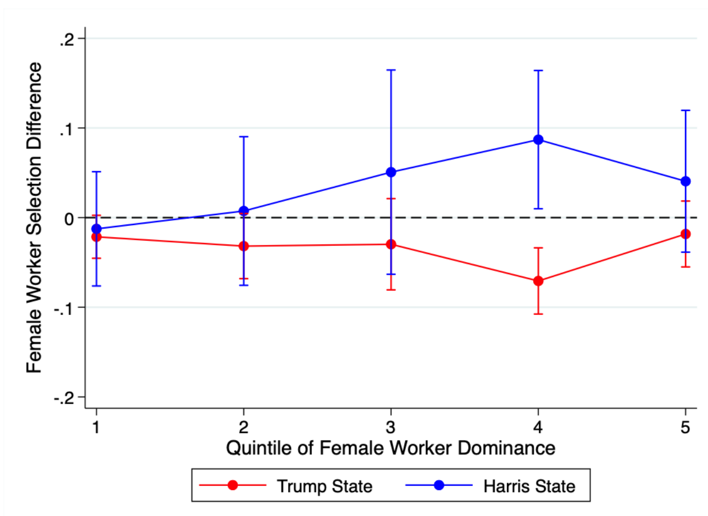
Figure B16: Female worker selection by state election winner,
by likelihood that female worker is more productive



(a) Baseline



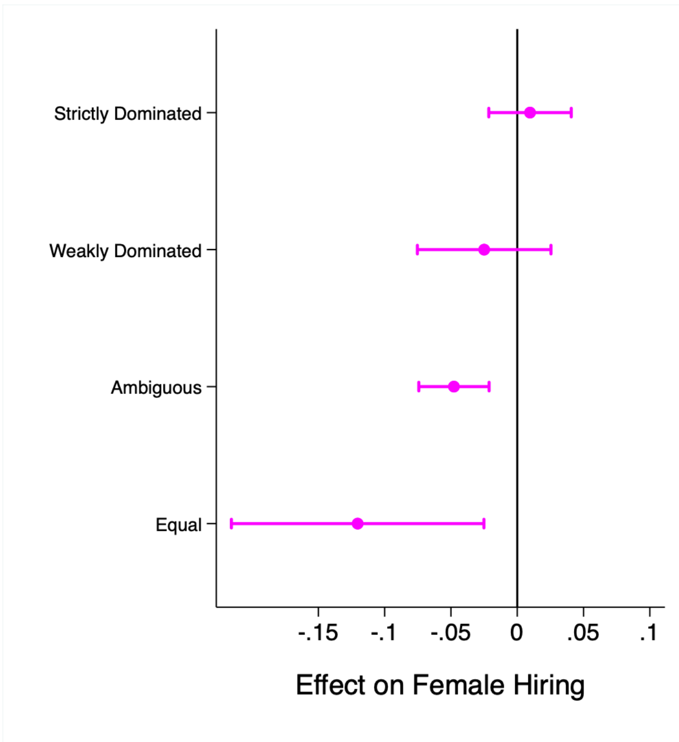
(b) Endline



(c) Difference (Endline minus Baseline)

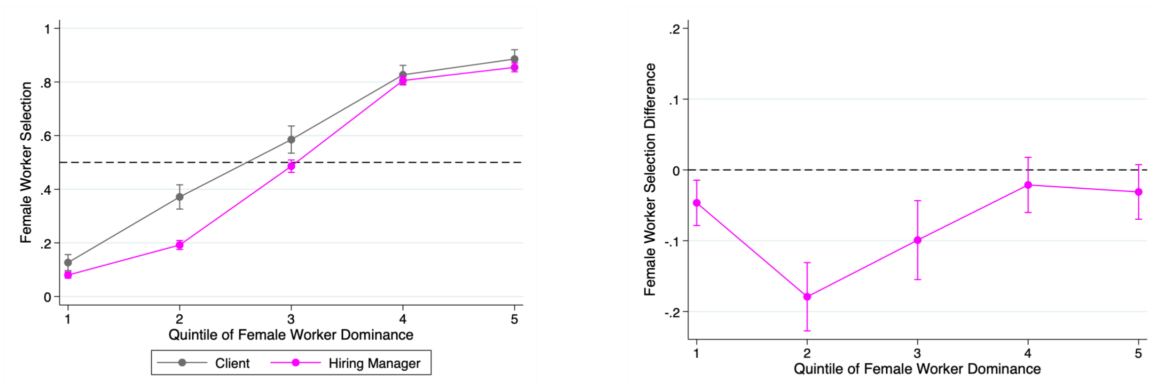
Notes: Panels (a) and (b) show the mean proportion of female workers hired by managers split (i) by whether Trump (red dots) vs. Harris (blue dots) won the highest vote share in the election in that manager's state, and (ii) across different subsets of mixed gender pairs based on the relative qualifications of female vs. male workers in each worker pair. The x-axis represents the quintile of relative female-male worker qualifications measured by the predicted likelihood that the female worker outperformed the male worker on Quiz B given their (i) highest level of education, and (ii) Quiz A score. Quintiles 1 and 2 include worker pairs where male workers were more qualified than female workers (i.e., predicted to outperform them on Quiz B), whereas quintiles 4 and 5 include worker pairs where female workers typically outperform male workers on Quiz B. Panel (a) includes results for the baseline survey and Panel (b) for the endline survey. Panel (c) shows the change in the proportion of female workers hired between baseline and endline broken down by election winner across different level of female-male worker qualification. Lines represent 95% confidence intervals of the means.

Figure B17: Hiring Manager vs Client Female Worker Selection by Type of Worker Pair



Notes: Each dot represents the regression estimated difference between hiring managers' and clients' likelihood of hiring a female worker. The coefficient is specifically on the variable $HiringManager_i$ from an OLS regression of the decision weight placed on the worker being female (Y_i, the dependent variable) on an indicator that the decision maker was a hiring manager ($HiringManager_i = 1$) rather than a client ($HiringManager_i = 0$). I pool both client and hiring manager observations from both baseline and endline surveys when running this regression, and run it separately for Y_i derived from decisions over different types of worker pairs, such as pairs where: (i) one worker strictly dominates (based on highest education and Quiz A score), (ii) one worker weakly dominates, (iii) one worker is higher on one signal and lower on the other, and (iv) where worker signals are equal. Lines represent 95% confidence intervals of the estimated coefficients.

Figure B18: Female worker selection by Hiring Manager vs. Client, by likelihood that female worker is more productive



(a) Hiring and Client

(b) Difference (Hiring minus Client)

Notes: Panel (a) shows the mean proportion of female workers selected by (i) hiring managers (magenta dots dots) vs. clients (grey dots), and (ii) across different subsets of mixed gender pairs based on the relative qualifications of female vs. male workers in each worker pair. The x-axis represents the quintile of relative female-male worker qualifications measured by the predicted likelihood that the female worker outperformed the male worker on Quiz B given their (i) highest level of education, and (ii) Quiz A score. Quintiles 1 and 2 include worker pairs where male workers were more qualified than female workers (i.e., predicted to outperform them on Quiz B), whereas quintiles 4 and 5 include worker pairs where female workers typically outperform male workers on Quiz B. Panel (b) shows the different in the proportion of female workers selected between clients and hiring managers across different level of female-male worker qualification. Lines represent 95% confidence intervals of the means.

C Natural Experiment - Additional Tables

Table C6: Hiring Manager Baseline Beliefs about Election Results

	State-level				National-level			
	All States	Harris-Likely States	Swing States	Trump-Likely States	All States	Harris-Likely States	Swing States	Trump-Likely States
Belief - which candidate would win (proportion)								
Harris victory	0.52	0.63	0.57	0.13	0.59	0.52	0.61	0.55
Trump Victory	0.47	0.38	0.43	0.85	0.40	0.46	0.38	0.43
Other candidate Victory	0.00	0.00	0.00	0.01	0.01	0.01	0.01	0.01
Margin of victory beliefs (% Vote Difference)								
Absolute	12.74	17.23	11.65	13.09	7.35	9.31	7.26	5.47
Net, in Trump's favor	-0.48	-2.53	-1.80	8.98	-1.52	0.35	-2.21	-0.09
Conditional on Trump Victory Belief	12.85	19.60	11.53	12.76	7.17	10.34	6.53	6.04
Conditional on Harris Victory Belief	12.65	15.80	11.74	15.22	7.48	8.38	7.73	5.00
Certainty of candidate choice (Likert scale, 1-7)								
All	4.90	5.07	4.69	5.36	4.66	4.84	4.64	4.53
Conditional on Trump Victory Belief	4.93	5.38	4.83	4.67	4.67	4.60	4.76	4.24
Conditional on Harris Victory Belief	4.92	5.26	4.77	5.27	4.67	4.71	4.71	4.37

Notes: This table summarizes hiring managers' beliefs about the election outcome at both the national-level and the state-level These were measured during the baseline survey (i.e., before the election). Beliefs include (i) which candidate would win (incentivized choice), (ii) the predicted winning margin of this candidate (given in percentages here; incentivized), (iii) their level of self-reported certainty that their prediction about the winning candidate was correct (from 1-completely unsure, to 7- completely certain; unincentivized).

Table C7: Regression Results for Hiring Manager Belief Changes (of Client Discrimination) by "Belief Shock"

	1	2	3	4	5	6	7	8
Trump won state	-0.062***	-0.059***	-0.059***	-0.057***		-0.059***	-0.061***	-0.061***
	(0.022)	(0.022)	(0.022)	(0.022)		(0.022)	(0.018)	(0.018)
Surprise: election winner	-0.051***	-0.05***	-0.05***	-0.045***	-0.049***	-0.045***	-0.051***	-0.049***
	(0.014)	(0.014)	(0.014)	(0.014)	(0.018)	(0.015)	(0.011)	(0.011)
Surprise: winning margin	-0.134***	-0.138***	-0.138***	-0.121**	-0.138***	-0.118**	-0.122***	-0.122***
	(0.05)	(0.051)	(0.051)	(0.051)	(0.053)	(0.051)	(0.04)	(0.041)
	Main	Control	Control + Productivity belief	Control + Politics	Control + State Fixed Effects	All controls	Baseline Belief	Baseline Belief + Controls

Notes: Each column and row correspond to a separate OLS regression specification, with coefficients on the primary independent variable reported and robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Each row represents regressions with a different set of independent variables, representing the β_1 coefficient in regression equations 1, 2, and 3 respectively (shown in the Supplement A.4). The dependent variable (Y_i in columns 1-6) is the change in hiring manager's incentivized beliefs about client discrimination, calculated by subtracting the baseline from the endline belief measure (of beliefs of the proportion of workers clients selected that are female). Control variables vary by specification as indicated, but the base controls (columns 2-6 and 8) include dummy variables for black and white race, male gender, college educated (or not), and continuous age and age squared. Productivity belief indicates incentivized beliefs about the relative male and female worker productivity, measured by predicted task scores; Politics refers to political party affiliation fixed effects, and "All Controls" represent all the above control variables, and additionally income and highest education level fixed effects. Columns 7 and 8 provide results for regressions using the alternative specification recommended by McKenzie (2012), which regresses endline beliefs on my key independent variables, controlling for baseline beliefs (rather than using the change in beliefs as the dependent variable).

Table C8: Hiring Manager Third-Party Discrimination Beliefs:
Client Hiring and General Others in the State,
Baseline and Endline Means by State Winner and Prediction

		Trump State		Harris State	
	Round	Trump Surprise	Trump Expected	Harris Expected	Harris Surprise
Client Female Hiring	Baseline	0.448***	0.453***	0.461*	0.43**
		(0.01)	(0.011)	(0.022)	(0.033)
	Endline	0.405***	0.452***	0.481	0.5
		(0.01)	(0.011)	(0.02)	(0.032)
Client Female Hiring Likert	Baseline	2.575***	2.576***	2.725**	3.192
		(0.062)	(0.062)	(0.124)	(0.184)
	Endline	2.31***	2.541***	2.675***	3.038
		(0.062)	(0.065)	(0.126)	(0.162)
Gallup out of 100 ”work out home”	Baseline	79.966	84.343	83.125	78.731
		(1.402)	(1.076)	(2.929)	(3.706)
	Endline	78.971	81.448	80.600	80.962
		(1.181)	(1.366)	(3.128)	(3.179)
WVS Likert: ”Jobs for men”	Baseline	4.678	4.680	4.575	4.462
		(0.13)	(0.138)	(0.299)	(0.377)
	Endline	4.19	4.715	4.875	4.962
		(0.129)	(0.132)	(0.254)	(0.326)
Observations		174	172	40	26

Notes:. Columns 2-5 display mean levels of hiring manager beliefs about third-party discrimination (both discrimination by clients and general others in their state) at both baseline and endline. Standard errors in parentheses. This is split by both states in which Trump won (columns 2 and 3) vs. Harris won the highest vote share (columns 4 and 5) and by the “surprisingness” of the result given participants' baseline prediction about which candidate would win the election (“Expected” indicates they correctly predicted the winner and “Surprised” that they failed to predict the winner). Stars represent statistical significance at the following thresholds *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ - for tests against benchmarks of neither favoring nor disfavoring female workers (only for Client Female Hiring and Client Female Hiring Likert, as there are no normative benchmarks for the other measures). These are 0.5 for Client Female Hiring, 3 for Client Female Hiring Likert. Higher values indicating more positive attitudes towards female workers.

Table C9: Hiring Manager Third-Party Discrimination Beliefs: Change (over Election) by State Winner and Prediction: Client Hiring and General Others in the State

	Trump State			Harris State		
	Trump Surprise	Trump Expected	Diff: Surprise - Expected	Harris Expected	Harris Surprise	Diff: Surprise - Expected
Change in Client Female Hiring	-0.043***	-0.001	-0.041**	0.021	0.07*	0.05
	(0.011)	(0.014)	(0.018)	(0.021)	(0.04)	(0.042)
Change in Client Female Hiring Likert	-0.264***	-0.035	-0.229**	-0.05	-0.154	-0.104
	(0.07)	(0.065)	(0.096)	(0.147)	(0.233)	(0.262)
Change in Gallup out of 100 ”work out home”	-0.994	-2.895**	1.901	-2.525	2.231	4.756
	(1.435)	(1.318)	(1.949)	(2.801)	(3.369)	(4.41)
Change in WVS Likert: ”Jobs for men”	-0.489***	0.035	-0.523***	0.3	0.5	0.2
	(0.141)	(0.133)	(0.194)	(0.218)	(0.339)	(0.384)
Observations	174	172		40	26	

Notes:. Columns 1-2 and 4-5 display mean changes over the election period (i.e., between baseline and endline surveys) in hiring manager beliefs about third-party discrimination. Standard errors in parentheses. This is split by both states in which Trump won (columns 1 and 2) vs. Harris won the highest vote share (columns 4 and 5) and by the “surprisingness” of the result given participants’ baseline prediction about which candidate would win the election (“Expected” indicates they correctly predicted the winner and “Surprised” that they failed to predict the winner). Stars represent statistical significance at the following thresholds *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ - for tests against 0, which represents no change, with higher values indicating more positive beliefs about attitudes towards female workers after the election. Columns 3 and 6 display the difference between values in columns 1 and 2 and between values in columns 5 and 4, respectively (i.e., “Surprise” columns minus “Expected” columns). Again, stars represent statistical significance at the following thresholds *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ - for tests against 0.

Table C10: Hiring Manager Female Worker Selection: Baseline and Endline Means by State Winner and Prediction

		Trump State		Harris State	
	Round	Trump Surprise	Trump Expected	Harris Expected	Harris Surprise
Proportion of Female Workers Hired	Baseline	0.494	0.484*	0.475	0.479
		(0.009)	(0.009)	(0.017)	(0.022)
	Endline	0.458***	0.45***	0.493	0.526
		(0.009)	(0.009)	(0.015)	(0.025)
Gender (Female) Weight	Baseline	0.008	-0.004	-0.006	-0.018
		(0.009)	(0.009)	(0.017)	(0.021)
	Endline	-0.028***	-0.032***	0.012	0.051**
		(0.01)	(0.011)	(0.017)	(0.024)
Education Weight	Baseline	0.088***	0.108***	0.072***	0.034*
		(0.012)	(0.012)	(0.023)	(0.02)
	Endline	0.105***	0.086***	0.089***	0.072***
		(0.012)	(0.011)	(0.025)	(0.024)
Quiz A Weight	Baseline	0.307***	0.317***	0.315***	0.297***
		(0.012)	(0.011)	(0.027)	(0.04)
	Endline	0.305***	0.32***	0.337***	0.32***
		(0.012)	(0.011)	(0.018)	(0.032)
Observations		174	172	40	26

Notes:. Columns 2-5 display mean levels of hiring manager female worker selection, including the proportion hired and the respective regression-estimated decision weights, at both baseline and endline. Standard errors in parentheses. Decision weights are estimated by regressing choices for each participant on the difference in gender, education signal, and Quiz A signal for each worker pair. These statistics are split by both states in which Trump won (columns 2 and 3) vs. Harris won the highest vote share (columns 4 and 5) and by the "surprisingness" of the result given participants' baseline prediction about which candidate would win the election ("Expected" indicates they correctly predicted the winner and "Surprised" that they failed to predict the winner). Stars represent statistical significance at the following thresholds *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ - for tests against a benchmark of 0.5 for the Proportion of Female Workers Hired and against 0 for all regression-estimated decision weights.

Table C11: Hiring Manager Female Worker Selection: Change (over Election) by State Winner and Prediction

	Trump State			Harris State		
	Trump Surprise	Trump Expected	Diff: Surprise - Expected	Harris Expected	Harris Surprise	Diff: Surprise - Expected
Proportion of Female Workers Hired	-0.036***	-0.034***	-0.002	0.018	0.046	0.028
	(0.011)	(0.011)	(0.015)	(0.019)	(0.03)	(0.034)
Gender (Female) Weight	-0.037***	-0.028***	-0.009	0.018	0.069**	0.051
	(0.011)	(0.011)	(0.015)	(0.018)	(0.028)	(0.031)
Education Weight	0.017	-0.023*	0.04*	0.017	0.038	0.02
	(0.013)	(0.013)	(0.021)	(0.023)	(0.028)	(0.036)
Quiz A Weight	-0.002	0.003	-0.005	0.022	0.023	0.001
	(0.014)	(0.012)	(0.018)	(0.025)	(0.043)	(0.046)
Observations	174	172		40	26	

Notes:. Columns 1-2 and 4-5 display mean changes over the election period (i.e., between baseline and endline surveys) in hiring manager female worker selection, including changes in the proportion hired and changes in the respective regression-estimated decision weights. Decision weights in each period are estimated by regressing choices for each participant on the difference in gender, education signal, and Quiz A signal for each worker pair. Standard errors of mean differences in parentheses. These statistics are split by both states in which Trump won (columns 1 and 2) vs. Harris won the highest vote share (columns 4 and 5) and by the "surprisingness" of the result given participants' baseline prediction about which candidate would win the election ("Expected" indicates they correctly predicted the winner and "Surprised" that they failed to predict the winner). Stars represent statistical significance at the following thresholds *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ - for tests against 0, which represents no change, with higher values indicating increases in these proportions or weights after the election. Columns 3 and 6 display the difference between values in columns 1 and 2 and between values in columns 5 and 4, respectively (i.e., "Surprise" columns minus "Expected" columns). Again, stars represent statistical significance at the following thresholds *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$ - for tests against 0.

Table C12: Regression Results for Client Choices and Beliefs, Regressed on State Election Winner

	Gender Weight in Choice					
	1	2	3	4	5	6
Trump won state	0.056 (0.062)	0.045 (0.063)	0.046 (0.058)	0.045 (0.059)	0.048 (0.057)	0.033 (0.058)
	Beliefs about Female-Male Productivity					
Trump won state	0.299 (0.727)	0.248 (0.734)	0.207 (0.728)	0.171 (0.732)	0.268 (0.39)	0.23 (0.367)
	Main	Control	Control + Politics	All controls	Baseline Control	Baseline Belief + Controls

Notes: Each column and row correspond to a separate OLS regression specification, with coefficients on the primary independent variable (an indicator for whether Trump won the highest vote share in the clients' state of residence) reported and robust standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. In the top panel, the dependent variable (in columns 1-4) is the change in the weight the client placed on gender (the weight placed on the worker being female) when making choices from before to after the election, calculated by subtracting the baseline from the endline gender weights, determined by clients' choice data. In the bottom panel, the dependent variable (in columns 1-4) is the change in clients' beliefs about female minus male scores on Quiz B (i.e., the change, over the election, in the predicted gender difference in actual productivity). This is calculated by the subtracting the baseline gender difference belief from the endline gender difference belief. Control variables vary by specification as indicated, but the base controls (columns 2-6) include dummy variables for black and white race, male gender, college educated (or not), and continuous age and age squared. Politics refers to political party affiliation fixed effects, and "All Controls" represent all the above control variables, and additionally income and highest education level fixed effects. Columns 5 and 6 provide results for regressions using the alternative specification recommended by McKenzie (2012), which regresses endline gender weight (top panel) and endline belief of gender-score differences (bottom panel) on the Trump State indicator, controlling for the baseline measure of each dependent variable, respectively (rather than using the *change* variables as the dependent variables).

D Controlled Experiment - Supplements

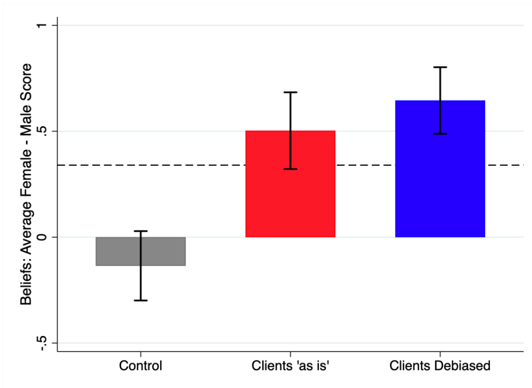
D.1 Replication of Belief Accuracy and Exacerbated Discrimination

This section presents the full results for the replication of the belief-accuracy analyses from the election natural experiment, using data from the controlled experiment. As preregistered, all analyses in this section use only participants assigned to the *Control* conditions (for both hiring managers and clients), since these groups' beliefs were not manipulated. This allows a test of how decision makers make these choices in the absence of (usually unavailable) information on productivity by gender, and a direct comparison to the election experiment, where no such information was provided.

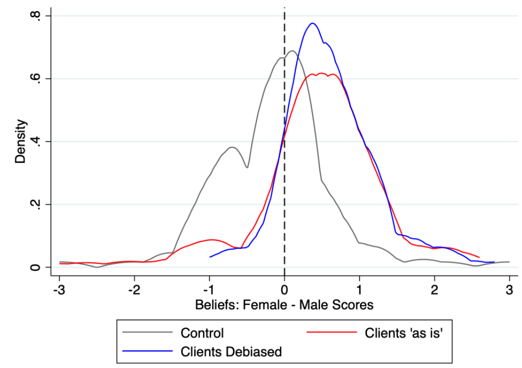
First, hiring managers significantly overestimated clients' discrimination against female workers: managers anticipated that clients would hire female workers 43.5% of the time, on average, 6.0 percentage points less than clients' actual female worker selection ($t = 5.11$, $CI = [0.036; 0.083]$, $p < 0.001$). This closely mirrors the overestimation observed in the election experiment (6.9 percentage points).

Second, hiring managers once again exacerbated discrimination in this labor market due to these inaccurate beliefs. I pool decision data for clients and hiring managers in the control groups and regress the choice to select a female worker in a given worker pair on a dummy variable for whether the decision maker was a hiring manager, controlling for worker signals; this constitutes the primary preregistered analysis for this hypothesis. Hiring managers were 5.4 percentage points less likely than clients to hire an equivalent female worker ($t = -2.91$, $CI = [-0.017; -0.091]$, $p = 0.004$; Figure E24), again closely mirroring the election experiment (6.1 percentage points).

Figure D19: Controlled Experiment: Hiring Manager Beliefs about Female-Male Worker Productivity, by Treatment Condition



(a) Means



(b) Kernel Density Functions

Notes: Panel (a) shows the mean difference between hiring managers' predicted Quiz B score for female workers and their predicted Quiz B score for male workers, in each treatment condition in the controlled experiment. The dotted line represents the actual difference (female workers scored on average 0.336 points more than male workers). Error bars represent 95% confidence intervals of the mean belief difference. Panel (b) represents the kernel density function of these belief differences by treatment condition.

D.2 Heterogeneity by Relative Worker Qualifications

Across 15 randomly selected worker pairs in the hiring task, there was substantial variation in workers' relative qualifications, or signals, in each pair (i.e., differences between workers' highest level of education and their scores on Quiz A). This allows me to test how the effect of third-party discrimination beliefs on female worker hiring differed as a function of workers' relative signals in each pair, e.g., in pairs where male workers were more qualified than female workers vs. pairs where female workers were more qualified. I investigate heterogeneity in the effect of third-party discrimination beliefs across two dimensions of worker pairs. First, as preregistered in the controlled experiment, I split worker pairs into categories based on the relative qualifications of workers in each pair, defining pairs where (i) one worker was better on at least one signal and no worse on the other (strictly and weakly dominated pairs), (ii) where one worker was better on one signal and worse on the other (ambiguous pairs), and (iii) where both workers had identical signals (equal pairs). This allows me to test whether changes in third-party beliefs affect female worker hiring when discrimination is more masked or *implicit*, as in ambiguous pairs, compared to where it is more transparent, as in dominated or equal pairs (Barron et al., 2024).[26] I find that

[26]Barron et al. (2024) use the method proposed by Cunningham and De Quidt (2022) to identify hiring discrimination as both implicit (using worker pairs as in the "ambiguous" case described in this manuscript), and explicit (using worker pairs where discrimination based on gender is transparent given other signals of worker productivity).

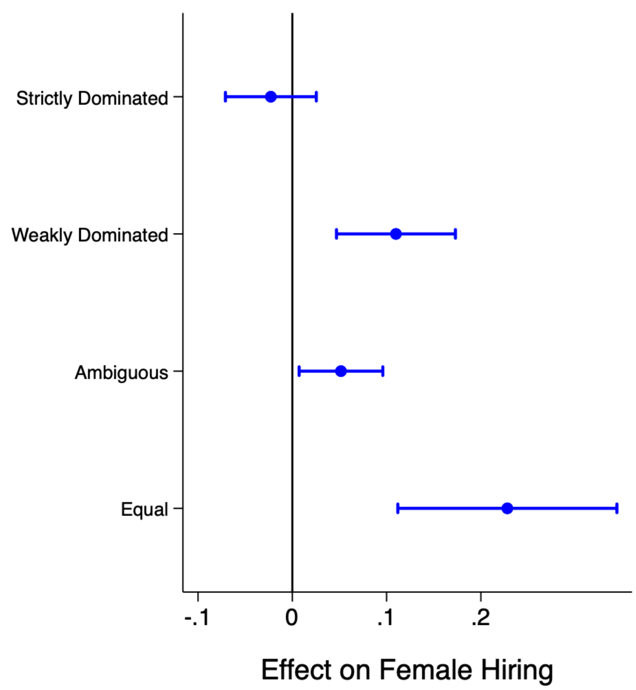
manipulating hiring manager beliefs about client choices had the strongest effect on choices when the pairs were *equal*: changing expected discrimination in the *Client Debiased* condition caused an increase in the likelihood of choosing a female worker of 22.8 percentage points in these pairs (Appendix Figure D20). This compares with an increase of 8.5 percentage points in ambiguous pairs and 9.8 percentage points in weakly dominated pairs. The effect for strictly dominated pairs was approximately zero.

Second, I divide worker pairs by the predicted productivity differences between female and male workers in each mixed-gender pair, given their signals of productivity (i.e., each workers' highest level of education and score on Quiz A). I used a logit model to predict the worker with the highest Quiz B across all worker pairs, using their highest level of education and score on Quiz A as predictors. I then used this model to compute the likelihood that the female worker in each mixed-gender pair had a higher Quiz B score, and divided worker pairs into quintiles based on this probability. I find that the *Clients Debiased* treatment had the largest effect when female and male workers were similarly qualified (i.e., they had similar predicted productivity given their qualifications). When I divide worker pairs into quintiles of relative female vs. male qualifications, I find the largest effect of the debiasing treatment in the 3rd quintile, where male and female workers were approximately equally qualified ($\beta = 0.175$, $t = 3.81$, $p < 0.001$; Appendix Figure D21). I find the smallest effects, neither of which are significant, in quintile 1 (where male workers are clearly more qualified) and quintile 5 (where female workers are clearly more qualified).

Taken together, it appears that manipulating decision maker beliefs about third-party discrimination had a larger effect on female worker hiring when workers were more similarly qualified, and worker gender could serve as a "tiebreaker".

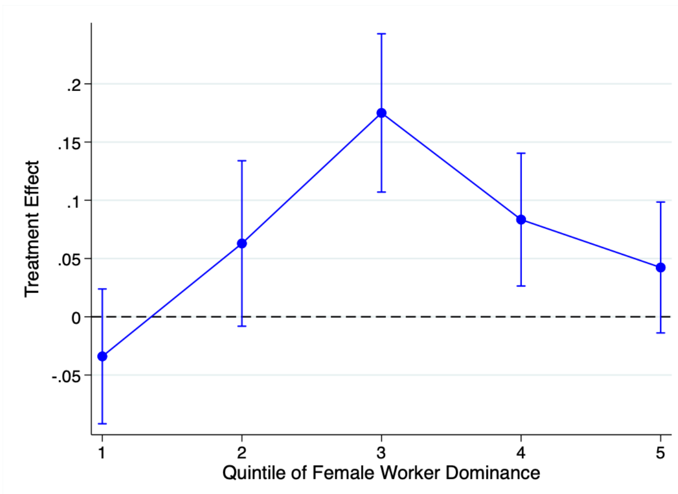
The authors find both forms of discrimination based on gender in hiring. This motivated this comparison in the present paper.

Figure D20: Controlled Experiment: Effect of *Clients Debiased* by Type of Worker Pair



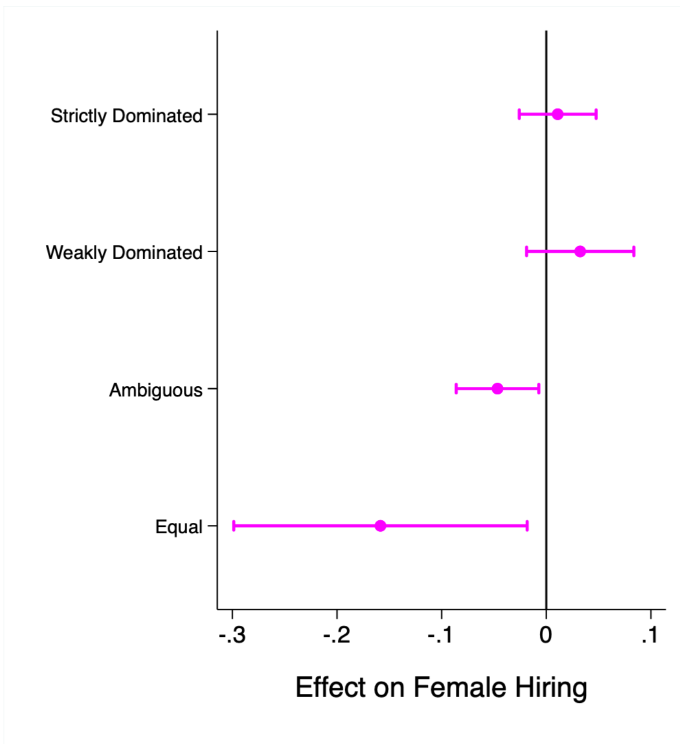
Notes: Each dot represents the coefficient on $ClientsDebiased_i$ from the following preregistered regression model (estimated using OLS), run on different sets of worker pairs from the hiring task: regressing a binary indicator for whether or not a hiring manager selected a female worker in a given worker pair on an indicator of whether the hiring manager was assigned to *either* the *Clients Debiased* condition or the *Clients as-is* condition ($GenderScoreInformation_i$; i.e., that they saw the worker gender score distribution), an indicator for whether the hiring manager was assigned to the *Clients Debiased* condition $ClientsDebiased_i$, and controlling for workers' relative scores on Quiz A and highest level of education, with standard errors clustered by participant. This regression is estimated separately for different worker pairs, where: (i) one worker strictly dominates (based on highest education and Quiz A score), (ii) one worker weakly dominates, (iii) one worker is higher on one signal and lower on the other, and (iv) where worker signals are equal. Lines represent 95% confidence intervals of the estimated coefficients.

Figure D21: Controlled Experiment: The Effect of Client-Debiased Treatment by likelihood that female worker is more productive



Notes: Blue dots show the coefficient on $ClientsDebiased_i$ from the following preregistered regression model (estimated using OLS), run on different sets of worker pairs from the hiring task: regressing a binary indicator for whether or not a hiring manager selected a female worker in a given worker pair on an indicator of whether the hiring manager was assigned to *either* the *Clients Debiased* condition or the *Clients as-is* condition ($GenderScoreInformation_i$; i.e., that they saw the worker gender score distribution), an indicator for whether the hiring manager was assigned to the *Clients Debiased* condition $ClientsDebiased_i$, and controlling for workers' relative scores on Quiz A and highest level of education, with standard errors clustered by participant. I run this regression separately across different subsets of mixed gender pairs based on the relative qualifications of female vs. male workers in each worker pair. The x-axis represents the quintile of relative female-male worker qualifications measured by the predicted likelihood that the female worker outperformed the male worker on Quiz B given their (i) highest level of education, and (ii) Quiz A score. Quintiles 1 and 2 include worker pairs where male workers were more qualified than female workers (i.e., predicted to outperform them on Quiz B), whereas quintiles 4 and 5 include worker pairs where female workers typically outperform male workers on Quiz B. Lines represent 95% confidence intervals of the estimated coefficients in each quintile.

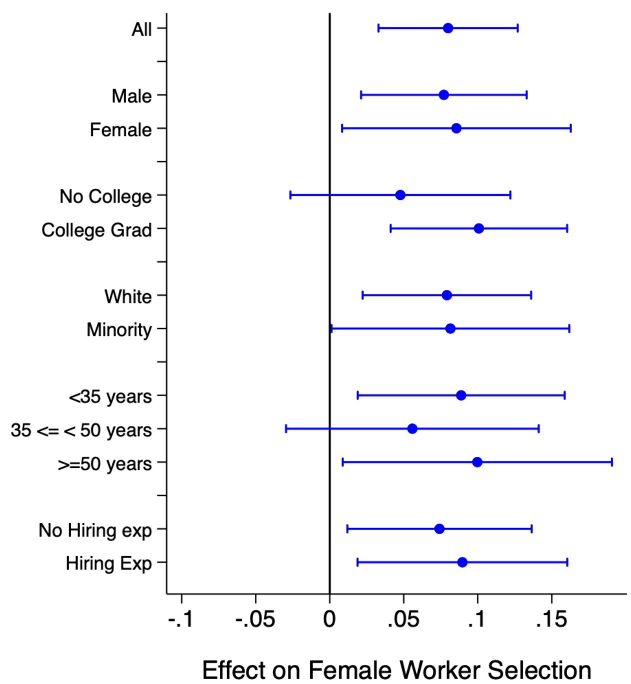
Figure D22: Controlled Experiment: Hiring Manager vs. Client Female Worker Selection by Type of Worker Pair



Notes: Each dot represents the regression estimated difference between hiring managers' and clients' likelihood of hiring a female worker in the *Control* condition. The coefficient is specifically on the variable $HiringManager_i$ from an OLS regression of the decision to hire a female worker in a given pair (Y_i, the dependent variable) on an indicator that the decision maker was a hiring manager ($HiringManager_i = 1$) rather than a client ($HiringManager_i = 0$), controlling for workers' relative scores on Quiz A and highest level of education, with standard errors clustered by participant. I run this regression on pooled observations from clients and hiring managers, but only those in the control groups. This is estimated separately for different worker pairs, such that: (i) one worker strictly dominates (based on highest education and Quiz A score), (ii) one worker weakly dominates, (iii) one worker is higher on one signal and lower on the other, and (iv) where worker signals are equal. Lines represent 95% confidence intervals of the estimated coefficients.

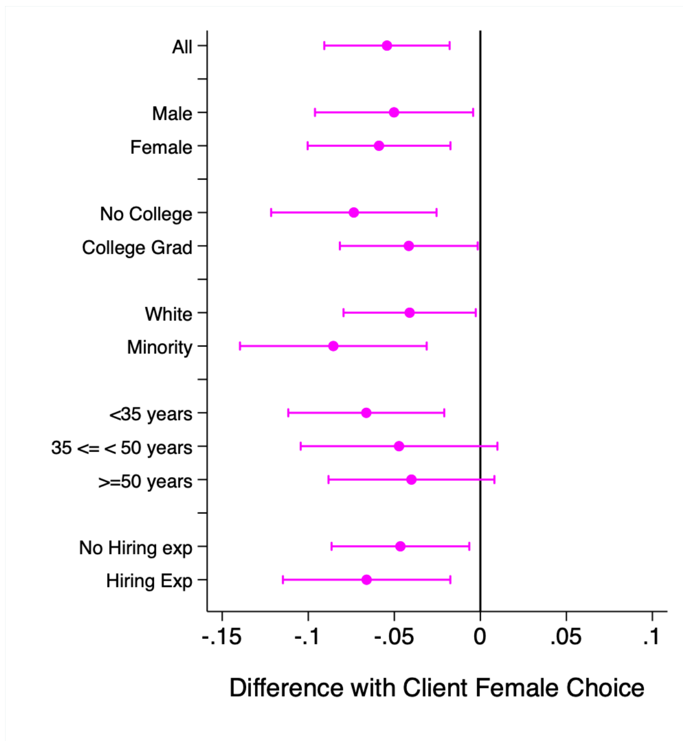
E Controlled Experiment - Additional Figures

Figure E23: Controlled Experiment: Heterogeneity of Effect of *Clients Debiased*



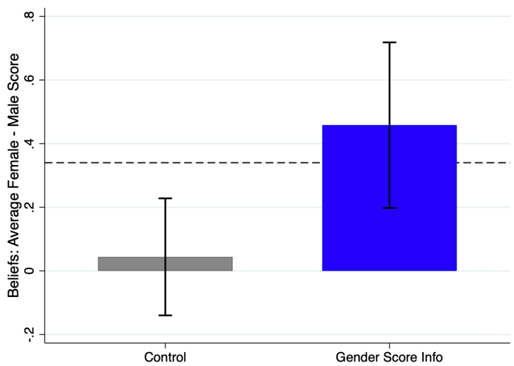
Notes: Each dot represents the coefficient on $ClientsDebiased_i$ from the following preregistered regression model (estimated using OLS), for different subgroups of hiring managers: regressing a binary indicator for whether or not a hiring manager selected a female worker in a given worker pair on an indicator of whether the hiring manager was assigned to *either* the *Clients Debiased* condition or the *Clients as-is* condition ($GenderScoreInformation_i$; i.e., that they saw the worker gender score distribution), an indicator for whether the hiring manager was assigned to the *Clients Debiased* condition $ClientsDebiased_i$, and controlling for workers' relative scores on Quiz A and highest level of education, with standard errors clustered by participant. The y-axis describes the specific subgroup of managers for which this regression was estimated, estimating this separately based on manager gender, highest level of education, race, age, and previous hiring experience respectively.

Figure E24: Controlled Experiment: Heterogeneity in Wedge between Hiring Manager and Client Female Worker Selection

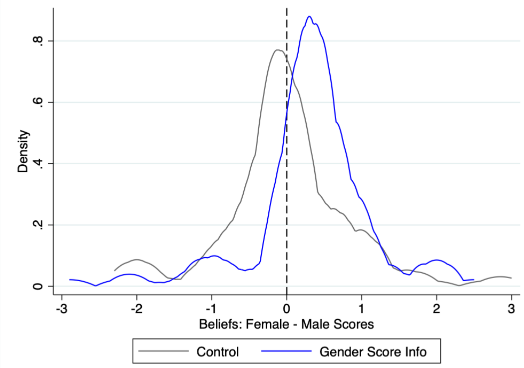


Notes: Each dot represents the coefficient on $HiringManager_i$ from the following preregistered regression model (estimated using OLS), for different subgroups of participants (i.e., hiring managers and clients): I pool observations (i.e., pair-level decisions in the hiring task) for both hiring managers and clients assigned to the *Control* condition, and regress a binary indicator for whether or not a participant selected a female worker in a given worker pair on an indicator of whether the participant was a hiring manager (as opposed to a client; $HiringManager_i$), and controlling for workers' relative scores on Quiz A and highest level of education, with standard errors clustered by participant. The y-axis describes the specific subgroup of managers for which this regression was estimated, estimating this separately based on manager gender, highest level of education, race, age, and previous hiring experience respectively.

Figure E25: Controlled Experiment: Client Beliefs about Female-Male Worker Productivity by Treatment Condition



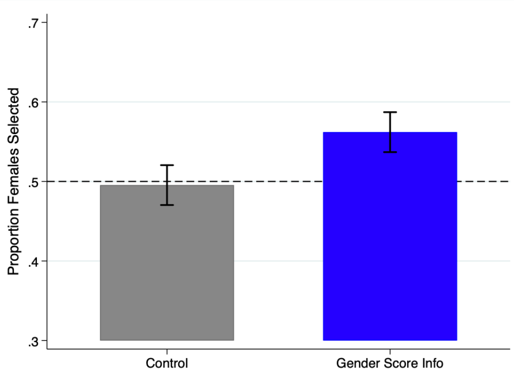
(a) Means



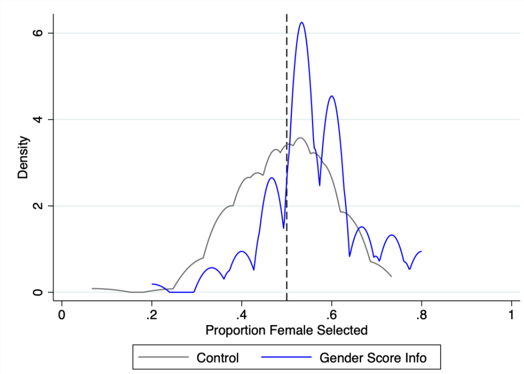
(b) Kernel Density Functions

Notes: Panel (a) shows the mean difference in clients' predicted Quiz B score for male workers minus the predicted Quiz B score for female workers, in each treatment condition in the controlled experiment. The dotted line represents the true difference of 0.337. Error bars represent 95% confidence intervals of the mean belief difference. Panel (b) represents the kernel density function of these prediction differences by treatment condition.

Figure E26: Controlled Experiment: Client Female Worker Selection by Treatment Condition



(a) Means



(b) Kernel Density Functions

Notes: Panel (a) shows the mean proportion of female workers clients selected, in each treatment condition in the controlled experiment. The dotted line represents equal proportions of male and female workers (i.e., 0.5). Error bars represent 95% confidence intervals of the mean belief. Panel (b) represents the kernel density function of these choices by treatment condition.

F Controlled Experiment - Additional Tables

Table F13: Controlled Experiment: Hiring Manager Sample Descriptive Statistics

		(1) Control		(2) Clients as-is		(3) Clients Debiased		T-test P-value	
Variable	N	Mean/SD	N	Mean/SD	N	Mean/SD	(1)-(2)	(1)-(3)	(2)-(3)
Age	91	41.16 (15.02)	114	39.70 (14.26)	95	40.20 (12.82)	0.48	0.64	0.79
Male	91	0.59 (0.49)	114	0.48 (0.50)	95	0.42 (0.50)	0.11	0.02**	0.38
White	91	0.70 (0.46)	114	0.64 (0.48)	95	0.64 (0.48)	0.34	0.38	0.98
Black or African-American	91	0.12 (0.33)	114	0.20 (0.40)	95	0.19 (0.39)	0.12	0.20	0.82
Other Race	91	0.19 (0.39)	114	0.18 (0.38)	95	0.19 (0.39)	0.83	0.96	0.79
College Graduate	91	0.63 (0.49)	114	0.59 (0.49)	95	0.65 (0.48)	0.58	0.71	0.34
Has Advanced Degree	91	0.15 (0.36)	114	0.18 (0.38)	95	0.21 (0.41)	0.68	0.32	0.52
Left-Right Political Scale	91	3.10 (1.46)	114	3.11 (1.75)	94	3.04 (1.70)	0.95	0.81	0.77
Has Real Hiring Experience	91	0.38 (0.49)	114	0.39 (0.49)	95	0.43 (0.50)	0.88	0.52	0.59
F-test of joint significance (F-stat)							0.63	0.22	0.98
F-test, number of observations							205	185	208

Notes: This table gives mean values (and standard deviations in brackets) for each variable for hiring managers separately in each treatment condition. The values displayed for t-tests and the F-test are p-values. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table F14: Controlled Experiment: Client Sample Descriptive Statistics

Variable	N	(1) Control Mean/SD	N	(2) Gender Score Info Mean/SD	T-test P-value (1)-(2)
Age	102	38.25	100	38.95	0.70
		(12.94)		(12.49)	
Male	102	0.46	100	0.49	0.68
		(0.50)		(0.50)	
White	102	0.74	100	0.76	0.69
		(0.44)		(0.43)	
Black or African-American	102	0.13	100	0.15	0.64
		(0.34)		(0.36)	
Other Race	102	0.15	100	0.10	0.31
		(0.36)		(0.30)	
College Graduate	102	0.64	100	0.53	0.12
		(0.48)		(0.50)	
Has Advanced Degree	102	0.21	100	0.08	0.01**
		(0.41)		(0.27)	
Left-Right Political Scale	101	3.25	97	3.48	0.32
		(1.72)		(1.63)	
Has Real Hiring Experience	102	0.47	100	0.47	0.99
		(0.50)		(0.50)	
F-test of joint significance (F-stat)					0.32
F-test, number of observations					198

Notes: This table gives mean values (and standard deviations in brackets) for each variable for clients separately for each treatment condition. The value displayed for t-tests and the F-test are p-values. ***, **, and * indicate significance at the 1, 5, and 10 percent critical level.

Table F15: Controlled Experiment: Regression Results: The Effect of *Clients Debiased* on Hiring Manager Beliefs about Client Choices, and on Hiring Choices

	Beliefs: Client Female Selection		Hiring Choices: Female Selection		
	(1)	(2)	(3)	(4)	(5)
Gender Score Information	0.053***	0.053***	0.054**	0.063***	0.061***
	(0.019)	(0.019)	(0.022)	(0.021)	(0.021)
Clients Debiased	0.085***	0.081***	0.084***	0.080***	0.075***
	(0.021)	(0.021)	(0.025)	(0.024)	(0.023)
Quiz A Score Difference				-0.183***	-0.183***
				(0.004)	(0.004)
Education Difference				-0.134***	-0.135***
				(0.009)	(0.009)
Conditioned on Signal	-	-		X	X
All Controls Included		X			X
Observations	300	299	3190	3190	3180

Notes: Columns 1 and 2 include the results of OLS regressions of hiring managers' incentivized beliefs about the proportion of female workers that clients selected (the dependent variable). Columns 3-5 include results of regressions of a binary variable of whether a female worker was hired the (dependent variable) in a given decision for a mixed-gender worker pair, with standard errors clustered at the hiring manager level. Across models, the dependent variables are regressed on a binary indicator for whether hiring managers saw workers' Quiz B score distribution, broken down by worker gender ("Gender Score Information"; this includes all managers assigned to either the *Clients as-is* or the *Clients Debiased* conditions), and on a binary indicator for whether managers were assigned to the *Clients Debiased* condition (the primary manipulation of beliefs about client discrimination). In columns 4 and 5, regressions include controls for workers' relative productivity signals in each pair (i.e., their relative highest education level and score on Quiz A; coefficients reported), and in columns 2 and 5, regressions include additional controls for manager race, gender, age, age squared, highest education level, and political ideology. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table F16: Controlled Experiment: Instrumental Variable Regression Results: Causal Effect of Third-Party Beliefs on Hiring Choices

	(1)	(2)	(3)	(4)	(5)	(6)
Beliefs: Client Female Selection (Incentivized)	1.045***	0.993***	0.968***			
	(0.361)	(0.343)	(0.340)			
Beliefs: Client Female Selection (Likert)				0.520***	0.494***	0.448***
				(0.151)	(0.146)	(0.136)
Gender Score Information	-0.005	0.006	0.008	0.007	0.018	0.022
	(0.041)	(0.039)	(0.037)	(0.032)	(0.031)	(0.029)
Quiz A Score Difference		-0.183***	-0.183***		-0.182***	-0.182***
		(0.004)	(0.004)		(0.004)	(0.004)
Education Difference		-0.135***	-0.135***		-0.133***	-0.134***
		(0.009)	(0.009)		(0.009)	(0.009)
Conditioned on Signal		X	X		X	X
All Controls Included			X			X
Belief type	Incentivized	Incentivized	Incentivized	Likert	Likert	Likert
Observations	3190	3190	3180	3190	3190	3180

Notes: All columns present regression results for two-stage-least squares (2SLS) regressions of whether a female worker was hired (the dependent variable) in a given decision for a mixed-gender worker pair, with standard errors clustered at the hiring manager level. Across all columns, I test the causal effect of hiring manager beliefs about client discrimination (i.e., female worker selection) on hiring managers' own discrimination, using random assignment to the *Clients Debiased* condition as an instrument for beliefs about client discrimination. In columns 1-3, I use hiring managers' incentivized beliefs about the proportion of female workers clients selected as the main independent variable (i.e., measure of beliefs about client discrimination). In columns 4-6 I use their response to a 5-point Likert question as a measure of beliefs about client discrimination. Across models, I control for whether hiring managers saw workers' Quiz B score distribution, broken down by worker gender ("Gender Score Information"; this includes all managers assigned to either the *Clients as-is* or the *Clients Debiased* conditions). In columns 2-3 and 5-6, regressions include controls for workers' relative productivity signals in each pair (i.e., their relative highest education level and score on Quiz A; coefficients reported), and in columns 3 and 6, regressions include additional controls for manager race, gender, age, age squared, highest education level, and political ideology. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.

Table F17: Controlled Experiment: Regression Results for Clients' Beliefs about Female-Male Worker Productivity and Choices

	Beliefs: Female - Male Scores		Choices: Female Offers		
	(1)	(2)	(3)	(4)	(5)
Gender Score Information	0.414**	0.421***	0.093***	0.094***	0.100***
	(0.163)	(0.158)	(0.021)	(0.020)	(0.020)
Quiz A Score Difference				-0.179***	-0.179***
				(0.005)	(0.005)
Education Difference				-0.118***	-0.119***
				(0.009)	(0.010)
Conditioned on Signal	-	-		X	X
All Controls Included		X			X
Observations	202	198	2158	2158	2114

Notes: Columns 1 and 2 include the results of OLS regressions of the mean difference between clients' predicted Quiz B score for female workers and their predicted Quiz B score for male workers (the dependent variable). Columns 3-5 include results of regressions of a binary variable of whether a female worker was selected in a given decision or worker pair (the dependent variable), with standard errors clustered at the client level. Across models, the dependent variables are regressed on a binary indicator for whether clients saw workers' Quiz B score distribution, broken down by worker gender (i.e., were assigned to the "Gender Score Information" condition). In columns 4 and 5, regressions include controls for workers' relative productivity signals in each pair (i.e., their relative highest education level and score on Quiz A; coefficients reported), and in columns 2 and 5, regressions include additional controls for client race, gender, age, age squared, highest education level, and political ideology. Standard errors in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.